FULL MOLECULAR HAMILTONIAN

From Classical to Quantum

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1 Classical Cartesian Hamiltonian

We have N_{nucl} nuclei labeled by j with masses M_j, charges $Z_j e$, positions $\mathbf{R}_j$, and conjugate momenta $\mathbf{P}_j$; and N_{elec} electrons labeled by i with mass m_e, charge $-e$, positions $\mathbf{r}_i$, and momenta $\mathbf{p}_i$. All vectors live in a space-fixed (SF) lab frame. Using Gaussian units and writing $R_{ij} = |\mathbf{R}_i - \mathbf{R}_j|$ etc.,

$$H = \underbrace{\sum_{j=1}^{N_{\text{nucl}}} \frac{\mathbf{P}_j^2}{2M_j}}_{T_N} + \underbrace{\sum_{i=1}^{N_{\text{elec}}} \frac{\mathbf{p}_i^2}{2m_e}}_{T_e} + V, \tag{1.1}$$

$$V = \frac{1}{4\pi\epsilon_0}\left(\frac{1}{2}\sum_{i\neq j}\frac{Z_i Z_j e^2}{R_{ij}} - \sum_{i,j}\frac{Z_j e^2}{|\mathbf{R}_j - \mathbf{r}_i|} + \frac{1}{2}\sum_{i\neq j}\frac{e^2}{r_{ij}}\right). \tag{1.2}$$

Canonical Poisson brackets $\{R_{j\alpha}, P_{k\beta}\} = \delta_{jk}\delta_{\alpha\beta}$, $\{r_{i\alpha}, p_{j\beta}\} = \delta_{ij}\delta_{\alpha\beta}$.

2 Transformation to the Nuclear Center-of-Mass

2.1 New coordinates

Define the nuclear COM, $3(N-1)$ internal nuclear vectors, and electron positions in the nuclear COM frame:

$$\mathbf{R}_{\text{cm}} = \frac{1}{M_N}\sum_j M_j \mathbf{R}_j, \qquad M_N = \sum_j M_j, \tag{2.1}$$

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\text{cm}}, \qquad \mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{cm}}. \tag{2.2}$$

The $\mathbf{R}'_j$ are linearly dependent:

$$\sum_j M_j \mathbf{R}'_j = 0. \tag{2.3}$$

2.2 Canonical momenta

This is a point transformation $\mathbf{Q}'_{\text{new}} = \mathbf{Q}'_{\text{new}}(\mathbf{q}_{\text{old}})$ with $\mathbf{q}_{\text{old}} = (\mathbf{R}_j, \mathbf{r}_i)$ and $\mathbf{Q}'_{\text{new}} = (\mathbf{R}_{\text{cm}}, \mathbf{R}'_j, \mathbf{r}'_i)$. Use the generating function: $F_2 = F_2(\mathbf{q}_{\text{old}}, \mathbf{P}, t) = \mathbf{P}\cdot\mathbf{Q}'_{\text{new}}(\mathbf{q}_{\text{old}})$

$$F_2 = \mathbf{P}_{\text{cm}}\cdot\mathbf{R}_{\text{cm}}(\{\mathbf{R}_j\}) + \sum_j \mathbf{P}'_j\cdot\mathbf{R}'_j(\{\mathbf{R}_j\}) + \sum_i \mathbf{p}'_i\cdot\mathbf{r}'_i(\mathbf{R}_j, \mathbf{r}_i). \tag{2.4}$$

Then $\mathbf{P}_j = \partial F_2/\partial\mathbf{R}_j$, $\mathbf{p}_i = \partial F_2/\partial\mathbf{r}_i$. Using

$$\frac{\partial\mathbf{R}_{\text{cm}}}{\partial\mathbf{R}_j} = \frac{M_j}{M_N}\mathbb{1}, \quad \frac{\partial\mathbf{R}'_k}{\partial\mathbf{R}_j} = \left(\delta_{jk} - \tfrac{M_j}{M_N}\right)\mathbb{1}, \quad \frac{\partial\mathbf{r}'_i}{\partial\mathbf{R}_j} = -\tfrac{M_j}{M_N}\mathbb{1}, \quad \frac{\partial\mathbf{r}'_i}{\partial\mathbf{r}_j} = \delta_{ij}\mathbb{1}, \tag{2.5}$$

one obtains

$$\mathbf{p}_i = \mathbf{p}'_i, \qquad \mathbf{P}_j = \frac{M_j}{M_N}\left(\mathbf{P}_{\text{cm}} - \textstyle\sum_i \mathbf{p}'_i - \sum_k \mathbf{P}'_k\right) + \mathbf{P}'_j. \tag{2.6}$$

The constraint dual to (2.3) is $\sum_k \mathbf{P}'_k = 0$, so

$$\boxed{\mathbf{P}_j = \frac{M_j}{M_N}\big(\mathbf{P}_{\rm cm} - \textstyle\sum_i \mathbf{p}_i\big) + \mathbf{P}'_j, \qquad \mathbf{p}_i = \mathbf{p}'_i} \tag{2.7}$$

Total momentum: $\sum_j \mathbf{P}_j + \sum_i \mathbf{p}_i = \mathbf{P}_{\rm cm}$. So $\mathbf{P}_{\rm cm}$ is the *total* (nuclei+electrons) momentum.

2.3 Hamiltonian after the transformation

Insert (2.7) into T_N in (1.1):

$$T_N = \sum_j \frac{\mathbf{P}_j^2}{2M_j} = \sum_j \frac{1}{2M_j}\left[\frac{M_j}{M_N}\big(\mathbf{P}_{\rm cm} - \textstyle\sum_i \mathbf{p}_i\big) + \mathbf{P}'_j\right]^2. \tag{2.8}$$

Expanding the square and summing,

$$T_N = \frac{(\mathbf{P}_{\rm cm} - \sum_i \mathbf{p}_i)^2}{2M_N} + \frac{1}{M_N}\big(\mathbf{P}_{\rm cm} - \sum_i \mathbf{p}_i\big)\cdot \underbrace{\sum_j \mathbf{P}'_j}_{=0} + \sum_j \frac{\mathbf{P}'^2_j}{2M_j}. \tag{2.9}$$

Combining with T_e and V:

$$H = \frac{\mathbf{P}_{\rm cm}^2}{2M_N} \underbrace{-\frac{\mathbf{P}_{\rm cm}\cdot\sum_i \mathbf{p}_i}{M_N}}_{\text{artifact}} + \underbrace{\frac{(\sum_i \mathbf{p}_i)^2}{2M_N}}_{\text{mass-polarization}} + \sum_j \frac{\mathbf{P}'^2_j}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + V(\mathbf{R}'_j, \mathbf{r}'_i). \tag{2.10}$$

Because H is independent of $\mathbf{R}_{\rm cm}$, $\mathbf{P}_{\rm cm}$ is conserved; pick the eigenvalue $\mathbf{P}_{\rm cm} = 0$ (center-of-mass at rest) and drop the trivial $\mathbf{P}_{\rm cm}$-pieces:

$$\boxed{H_{\rm int} = \sum_j \frac{\mathbf{P}'^2_j}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + \underbrace{\frac{1}{2M_N}\Big(\sum_i \mathbf{p}_i\Big)^2}_{\text{electron mass polarization}} + V(\mathbf{R}'_j, \mathbf{r}'_i).} \tag{2.11}$$

The potential is unchanged in form because $\mathbf{R}'_j - \mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_j$, $\mathbf{R}'_j - \mathbf{r}'_i = \mathbf{R}_j - \mathbf{r}_i$, $\mathbf{r}'_i - \mathbf{r}'_j = \mathbf{r}_i - \mathbf{r}_j$. The mass-polarization term, $(M_N)^{-1}\sum_{i<j}\mathbf{p}_i\cdot\mathbf{p}_j + (2M_N)^{-1}\sum_i \mathbf{p}_i^2$, is the price of referring electrons to the nuclear (rather than the total) COM. This rest-frame choice is not essential: Section 2.4 shows that the *same* internal Hamiltonian follows for any $\mathbf{P}_{\rm cm}$ from a classical canonical transformation, and it is that frame-independent form we carry forward.

2.4 Eliminating the Cross-Coupling via a Classical Canonical Transformation

The rest-frame choice $\mathbf{P}_{\rm cm} = 0$ made in Section 2.3 is convenient but stronger than necessary, and we now lift it. For $\mathbf{P}_{\rm cm} \neq 0$ the Hamiltonian (2.10) carries two artifacts inherited from having referred electrons to the *nuclear* (rather than the *total*) center of mass:

$$H = \underbrace{\frac{\mathbf{P}_{\mathrm{cm}}^{2}}{2M_N}}_{(A)\text{ wrong total mass}} \underbrace{-\frac{\mathbf{P}_{\mathrm{cm}}\cdot\sum_i \mathbf{p}'_i}{M_N}}_{(B)\text{ cross-coupling}} + \frac{(\sum_i \mathbf{p}'_i)^2}{2M_N} + \sum_j \frac{\mathbf{P}'^{\,2}_j}{2M_j} + \sum_i \frac{\mathbf{p}'^{\,2}_i}{2m_e} + V. \tag{2.12}$$

The wrong-mass term (A) has denominator M_N instead of the proper $M_{\mathrm{tot}} = M_N + N_{\mathrm{elec}} m_e$, and (B) couples the COM motion to the internal electronic momenta. We eliminate both *exactly, for any* $\mathbf{P}_{\mathrm{cm}}$, by a single classical canonical transformation generated on phase space — so that the internal Hamiltonian carried into the body-fixed treatment of Section 3 is the frame-independent object, not a rest-frame special case. This transformation is the classical preimage of the Galilean-boost unitary that one would apply in the quantum theory.

2.4.1 Generator

Define on phase space the function

$$G \equiv \frac{m_e}{M_{\mathrm{tot}}}\,\mathbf{P}_{\mathrm{cm}}\cdot\sum_i \mathbf{r}'_i, \qquad M_{\mathrm{tot}} = M_N + N_{\mathrm{elec}} m_e. \tag{2.13}$$

G depends only on $\mathbf{P}_{\mathrm{cm}}$ and $\{\mathbf{r}'_i\}$; in particular it is independent of $\mathbf{R}_{\mathrm{cm}}$, $\mathbf{R}'_j$, $\mathbf{P}'_j$, and $\mathbf{p}'_i$.

2.4.2 Induced canonical transformation

The infinitesimal canonical transformation generated by G acts on any phase-space function f by $\delta f = \{f, G\}$. Because G is linear in $\mathbf{P}_{\mathrm{cm}}$ and $\{\mathbf{r}'_i\}$, the bracket $\{\cdot, G\}$ applied twice annihilates the canonical variables, so the finite transformation at unit parameter is given by a single application:

$$\tilde{f} = f + \{f, G\}. \tag{2.14}$$

Using $\{f, g\} = \sum(\partial_q f\,\partial_p g - \partial_p f\,\partial_q g)$, compute the action on each canonical pair.
Electron pair $(\mathbf{r}'_i, \mathbf{p}'_i)$:

$$\begin{aligned} \{r'_{i,a}, G\} &= \tfrac{\partial G}{\partial p'_{i,a}} = 0, \\ \{p'_{i,a}, G\} &= -\tfrac{\partial G}{\partial r'_{i,a}} = -\tfrac{m_e}{M_{\mathrm{tot}}} P_{\mathrm{cm},a}. \end{aligned}$$

Nuclear-COM pair $(\mathbf{R}_{\mathrm{cm}}, \mathbf{P}_{\mathrm{cm}})$:

$$\begin{aligned} \{R_{\mathrm{cm},a}, G\} &= \tfrac{\partial G}{\partial P_{\mathrm{cm},a}} = \tfrac{m_e}{M_{\mathrm{tot}}} \sum_i r'_{i,a}, \\ \{P_{\mathrm{cm},a}, G\} &= -\tfrac{\partial G}{\partial R_{\mathrm{cm},a}} = 0. \end{aligned}$$

Internal nuclear pair $(\mathbf{R}'_j, \mathbf{P}'_j)$: G depends on neither, so $\{\mathbf{R}'_j, G\} = \{\mathbf{P}'_j, G\} = 0$. Hence the finite canonical transformation reads

$$\boxed{\begin{aligned} \tilde{\mathbf{r}}'_i &= \mathbf{r}'_i, & \tilde{\mathbf{p}}'_i &= \mathbf{p}'_i - \tfrac{m_e}{M_{\mathrm{tot}}}\mathbf{P}_{\mathrm{cm}}, \\ \tilde{\mathbf{R}}_{\mathrm{cm}} &= \mathbf{R}_{\mathrm{cm}} + \tfrac{m_e}{M_{\mathrm{tot}}}\textstyle\sum_i \mathbf{r}'_i, & \tilde{\mathbf{P}}_{\mathrm{cm}} &= \mathbf{P}_{\mathrm{cm}}, \\ \tilde{\mathbf{R}}'_j &= \mathbf{R}'_j, & \tilde{\mathbf{P}}'_j &= \mathbf{P}'_j. \end{aligned}} \tag{2.15}$$

2.4.3 Canonicity check

The only non-obvious Poisson bracket is between the two simultaneously shifted variables:

$$
\begin{aligned}
\{\tilde{p}'_{i,a}, \tilde{R}_{\mathrm{cm},b}\} &= \{\mathbf{p}'_{i,a} - \tfrac{m_e}{M_{\mathrm{tot}}} P_{\mathrm{cm},a},\ R_{\mathrm{cm},b} + \tfrac{m_e}{M_{\mathrm{tot}}} \textstyle\sum_l r'_{l,b}\} \\
&= \underbrace{\{p'_{i,a}, R_{\mathrm{cm},b}\}}_{0} + \tfrac{m_e}{M_{\mathrm{tot}}} \sum_l \underbrace{\{p'_{i,a}, r'_{l,b}\}}_{-\delta_{il}\delta_{ab}} \\
&\quad - \tfrac{m_e}{M_{\mathrm{tot}}} \underbrace{\{P_{\mathrm{cm},a}, R_{\mathrm{cm},b}\}}_{-\delta_{ab}} - \tfrac{m_e^2}{M_{\mathrm{tot}}^2} \sum_l \underbrace{\{P_{\mathrm{cm},a}, r'_{l,b}\}}_{0} \\
&= -\tfrac{m_e}{M_{\mathrm{tot}}} \delta_{ab} + \tfrac{m_e}{M_{\mathrm{tot}}} \delta_{ab} = 0.
\end{aligned}
\tag{2.16}
$$

All other elementary brackets are preserved by inspection (each is either unchanged or trivially zero), so (2.15) is canonical.

2.4.4 Transformed Hamiltonian

Invert (2.15) for substitution into (2.12): $\mathbf{p}'_i = \tilde{\mathbf{p}}'_i + (m_e/M_{\mathrm{tot}})\mathbf{P}_{\mathrm{cm}}$. The potential V depends only on $\mathbf{R}'_j$ and $\mathbf{r}'_i$, which are unchanged, so V is form-invariant. Working term by term.

Cross-coupling (B):

$$
\begin{aligned}
-\frac{\mathbf{P}_{\mathrm{cm}} \cdot \sum_i \mathbf{p}'_i}{M_N} &= -\frac{\mathbf{P}_{\mathrm{cm}} \cdot \left(\sum_i \tilde{\mathbf{p}}'_i + N_{\mathrm{elec}} \frac{m_e}{M_{\mathrm{tot}}} \mathbf{P}_{\mathrm{cm}}\right)}{M_N} \\
&= -\frac{\mathbf{P}_{\mathrm{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i}{M_N} - \frac{N_{\mathrm{elec}} m_e \mathbf{P}_{\mathrm{cm}}^2}{M_N M_{\mathrm{tot}}}.
\end{aligned}
\tag{2.17}
$$

Mass-polarization quadratic:

$$
\begin{aligned}
\frac{(\sum_i \mathbf{p}'_i)^2}{2M_N} &= \frac{\left(\sum_i \tilde{\mathbf{p}}'_i + N_{\mathrm{elec}} \frac{m_e}{M_{\mathrm{tot}}} \mathbf{P}_{\mathrm{cm}}\right)^2}{2M_N} \\
&= \frac{(\sum_i \tilde{\mathbf{p}}'_i)^2}{2M_N} + \frac{N_{\mathrm{elec}} m_e \mathbf{P}_{\mathrm{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i}{M_N M_{\mathrm{tot}}} + \frac{N_{\mathrm{elec}}^2 m_e^2 \mathbf{P}_{\mathrm{cm}}^2}{2M_N M_{\mathrm{tot}}^2}.
\end{aligned}
\tag{2.18}
$$

Electron kinetic energy:

$$
\begin{aligned}
\sum_i \frac{\mathbf{p}_i'^{\,2}}{2m_e} &= \sum_i \frac{(\tilde{\mathbf{p}}'_i + \frac{m_e}{M_{\mathrm{tot}}} \mathbf{P}_{\mathrm{cm}})^2}{2m_e} \\
&= \sum_i \frac{\tilde{\mathbf{p}}_i'^{\,2}}{2m_e} + \frac{\mathbf{P}_{\mathrm{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i}{M_{\mathrm{tot}}} + \frac{N_{\mathrm{elec}} m_e \mathbf{P}_{\mathrm{cm}}^2}{2M_{\mathrm{tot}}^2}.
\end{aligned}
\tag{2.19}
$$

Coefficient of $\mathbf{P}_{\mathrm{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i$*:* Summing (2.17), (2.18), (2.19),

$$
-\frac{1}{M_N} + \frac{N_{\mathrm{elec}} m_e}{M_N M_{\mathrm{tot}}} + \frac{1}{M_{\mathrm{tot}}} = \frac{-M_{\mathrm{tot}} + N_{\mathrm{elec}} m_e + M_N}{M_N M_{\mathrm{tot}}} = 0,
\tag{2.20}
$$

using $M_{\mathrm{tot}} = M_N + N_{\mathrm{elec}} m_e$. *The cross-coupling (B) is killed identically.*

Coefficient of $\mathbf{P}_{\mathrm{cm}}^2$*:* Adding the $\mathbf{P}_{\mathrm{cm}}^2/(2M_N)$ from (A) plus the residual $\mathbf{P}_{\mathrm{cm}}^2$-pieces from

(2.17)–(2.19),

$$
\begin{aligned}
\frac{1}{2M_N} - \frac{N_{\text{elec}} m_e}{M_N M_{\text{tot}}} + \frac{N_{\text{elec}}^2 m_e^2}{2M_N M_{\text{tot}}^2} + \frac{N_{\text{elec}} m_e}{2M_{\text{tot}}^2} &= \frac{1}{2M_N} - \frac{N_{\text{elec}} m_e}{M_N M_{\text{tot}}} + \frac{N_{\text{elec}} m_e}{2M_{\text{tot}}^2}\left(\frac{N_{\text{elec}} m_e}{M_N} + 1\right) \\
&= \frac{1}{2M_N} - \frac{N_{\text{elec}} m_e}{M_N M_{\text{tot}}} + \frac{N_{\text{elec}} m_e}{2M_{\text{tot}}^2} \cdot \frac{M_{\text{tot}}}{M_N} \\
&= \frac{1}{2M_N} - \frac{N_{\text{elec}} m_e}{M_N M_{\text{tot}}} + \frac{N_{\text{elec}} m_e}{2M_N M_{\text{tot}}} \\
&= \frac{1}{2M_N} - \frac{N_{\text{elec}} m_e}{2M_N M_{\text{tot}}} = \frac{M_{\text{tot}} - N_{\text{elec}} m_e}{2M_N M_{\text{tot}}} = \frac{M_N}{2M_N M_{\text{tot}}} = \frac{1}{2M_{\text{tot}}}.
\end{aligned}
\tag{2.21}
$$

The wrong-mass (A) is corrected to M_{tot}.
Assembling everything,

$$
\boxed{\tilde{H} = \frac{\mathbf{P}_{\text{cm}}^2}{2M_{\text{tot}}} \;+\; \sum_j \frac{\mathbf{P}_j^{\prime\,2}}{2M_j} \;+\; \sum_i \frac{\tilde{\mathbf{p}}_i^{\prime\,2}}{2m_e} \;+\; \frac{1}{2M_N}\Big(\sum_i \tilde{\mathbf{p}}_i'\Big)^2 \;+\; V(\mathbf{R}_j', \mathbf{r}_i').} \tag{2.22}
$$

The COM kinetic energy carries the *correct* total mass M_{tot}. The internal piece is structurally identical to (2.11): the same mass-polarization $(\sum_i \tilde{\mathbf{p}}_i')^2/(2M_N)$, the same nuclear kinetic energy, the same potential. Since $\tilde{H}$ is independent of $\tilde{\mathbf{R}}_{\text{cm}}$, $\tilde{\mathbf{P}}_{\text{cm}} = \mathbf{P}_{\text{cm}}$ is conserved, and one may proceed with the internal dynamics for *any* value of $\mathbf{P}_{\text{cm}}$ — not just zero — with no kinetic artifact.

2.4.5 Geometric interpretation

The new "COM coordinate" is

$$
\tilde{\mathbf{R}}_{\text{cm}} = \mathbf{R}_{\text{cm}} + \frac{m_e}{M_{\text{tot}}} \sum_i \mathbf{r}_i' = \frac{M_N \mathbf{R}_{\text{cm}} + m_e \sum_i \mathbf{r}_i}{M_{\text{tot}}} = \mathbf{R}_{\text{tot}}^{\text{CoM}}, \tag{2.23}
$$

using $\mathbf{r}_i = \mathbf{R}_{\text{cm}} + \mathbf{r}_i'$. The canonical transformation has *quietly relabeled* the COM coordinate from nuclear to total COM, while *leaving the relative coordinates* $\mathbf{R}_j'$ *and* $\mathbf{r}_i'$ *untouched*. This is precisely what makes the framework still "nuclear-COM-based": the Born–Oppenheimer-friendly relative coordinates of Section 2.1 survive unchanged. The momentum shift $\tilde{\mathbf{p}}_i' = \mathbf{p}_i' - (m_e/M_{\text{tot}})\mathbf{P}_{\text{cm}}$ is the kinetic compensation required to keep the Poisson brackets canonical: in Galilean terms, each electron's lab-frame momentum is reduced by $m_e \mathbf{V}_{\text{CoM}}$ with $\mathbf{V}_{\text{CoM}} = \mathbf{P}_{\text{cm}}/M_{\text{tot}}$.

The canonical transformation has thus cleanly separated the total COM motion *exactly and for any* $\mathbf{P}_{\text{cm}}$: in (2.22) the COM appears only through the decoupled free term $\mathbf{P}_{\text{cm}}^2/2M_{\text{tot}}$ — carrying the *correct* total mass M_{tot} and free of the cross-coupling (B) — while every other term depends only on the internal variables. It is this total Hamiltonian (2.22), exact for *any* $\mathbf{P}_{\text{cm}}$ and not merely the rest frame, that we carry into the body-fixed treatment of Section 3.

3 BF frame for both nuclei and electrons

3.1 Setup

With Eckart's frame fixed by the *nuclei*,

$$\mathbf{R}'_j = \mathbf{S}(\phi,\theta,\chi)\,\bar{\mathbf{R}}_j(Q), \qquad \mathbf{r}'_i = \mathbf{S}(\phi,\theta,\chi)\,\bar{\mathbf{r}}_i. \tag{3.1}$$

The Eckart conditions $\sum_j M_j \bar{\mathbf{R}}_j = 0$ and $\sum_j M_j \bar{\mathbf{R}}_j^{\rm eq} \times \bar{\mathbf{R}}_j = 0$ fix the orientation of $\mathbf{S}$ from the *nuclear* positions; the same $\mathbf{S}$ is then used to define BF electron coordinates $\bar{\mathbf{r}}_i$. The nuclear shape is parametrized by rectilinear mass-weighted **normal coordinates** Q_b,

$$\bar{\mathbf{R}}_j(Q) = \bar{\mathbf{R}}_j^{\rm eq} + \frac{1}{\sqrt{M_j}} \sum_b \mathbf{l}_{jb}\, Q_b, \tag{3.1a}$$

with mass-weighted normal-mode vectors $\mathbf{l}_{jb}$ obeying orthonormality and the Eckart (Sayvetz) conditions

$$\sum_j \mathbf{l}_{ja} \cdot \mathbf{l}_{jb} = \delta_{ab}, \qquad \sum_j \sqrt{M_j}\, \mathbf{l}_{jb} = 0, \qquad \sum_j \sqrt{M_j}\, \bar{\mathbf{R}}_j^{\rm eq} \times \mathbf{l}_{jb} = 0, \tag{3.1b}$$

so that $\partial \bar{\mathbf{R}}_j / \partial Q_b = \mathbf{l}_{jb}/\sqrt{M_j}$ is constant.
BF angular velocity:

$$(\mathbf{S}^T \dot{\mathbf{S}})_{\alpha\beta} = -\epsilon_{\alpha\beta\gamma}\, \omega_\gamma \Longrightarrow \mathbf{S}^T \dot{\mathbf{S}}\, \mathbf{v} = \boldsymbol{\omega} \times \mathbf{v}. \tag{3.2}$$

Velocities (using $|\mathbf{S}\mathbf{v}|^2 = |\mathbf{v}|^2$):

$$\dot{\mathbf{R}}'_j = \mathbf{S}\big(\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j\big), \qquad \dot{\mathbf{r}}'_i = \mathbf{S}\big(\boldsymbol{\omega} \times \bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i\big). \tag{3.3}$$

3.2 From the internal Hamiltonian back to a working Lagrangian

I need an action principle for the BF coordinates. I therefore Legendre-transform the total Hamiltonian $\tilde{H}$ of (2.22) — keeping $\mathbf{P}_{\rm cm}$ arbitrary, never setting it to zero — back to the velocity-form Lagrangian. The total COM coordinate $\tilde{\mathbf{R}}_{\rm cm}$ is cyclic, so its conjugate $\mathbf{P}_{\rm cm}$ is conserved. From Hamilton's equations applied to (2.22),

$$\dot{\tilde{\mathbf{R}}}_{\rm cm} = \mathbf{P}_{\rm cm}/M_{\rm tot}, \qquad \dot{\mathbf{R}}'_j = \mathbf{P}'_j/M_j, \qquad \dot{\mathbf{r}}'_i = \tilde{\mathbf{p}}'_i/m_e + (\textstyle\sum_l \tilde{\mathbf{p}}'_l)/M_N. \tag{3.4}$$

Summing the second: $\sum_i \dot{\mathbf{r}}'_i = (\sum_l \tilde{\mathbf{p}}'_l)\big(1/m_e + N_{\rm elec}/M_N\big) = (\sum_l \tilde{\mathbf{p}}'_l) \cdot M_{\rm tot}/(m_e M_N)$. Hence

$$\sum_l \tilde{\mathbf{p}}'_l = \frac{m_e M_N}{M_{\rm tot}} \sum_i \dot{\mathbf{r}}'_i, \qquad \tilde{\mathbf{p}}'_i = m_e \dot{\mathbf{r}}'_i - \frac{m_e^2}{M_{\rm tot}} \sum_l \dot{\mathbf{r}}'_l. \tag{3.5}$$

The total Lagrangian $\mathcal{L} = \sum p\,\dot{q} - \tilde{H}$ becomes

$$\boxed{\mathcal{L} = \frac{1}{2} M_{\rm tot} |\dot{\tilde{\mathbf{R}}}_{\rm cm}|^2 + \frac{1}{2} \sum_j M_j |\dot{\mathbf{R}}'_j|^2 + \frac{m_e}{2} \sum_i |\dot{\mathbf{r}}'_i|^2 - \frac{m_e^2}{2M_{\rm tot}} \Big|\sum_l \dot{\mathbf{r}}'_l\Big|^2 - V.} \tag{3.6}$$

(Verification of the electronic sector, $N_{\rm elec}=1$: its contribution to $\mathcal{L}$ is $\frac{1}{2}(m_e - m_e^2/M_{\rm tot})|\dot{\mathbf{r}}'_1|^2 = \frac{1}{2}\mu_e|\dot{\mathbf{r}}'_1|^2$, $\mu_e = m_e M_N/M_{\rm tot}$ — the correct hydrogenic reduced mass. ✓)

The mass polarization enters the Lagrangian with a negative sign — this is the crucial point. It is mathematically inevitable: the Legendre transform of a positive mass-polarization quadratic in momenta is a negative quadratic in velocities (cf. the sign-flip of a kinetic kernel under Legendre).

3.3 BF Lagrangian

Substitute (3.3) into (3.6); the COM term $\frac{1}{2}M_{\rm tot}|\dot{\bar{\mathbf{R}}}_{\rm cm}|^2$ is untouched by the body-fixed rotation. Define $\bar{\mathbf{r}} \equiv \sum_i \bar{\mathbf{r}}_i$. Since $\sum_l \dot{\mathbf{r}}'_l = \mathbf{S}(\boldsymbol{\omega}\times\bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}})$,

$$\mathcal{L} = \tfrac{1}{2}M_{\rm tot}|\dot{\bar{\mathbf{R}}}_{\rm cm}|^2 + \tfrac{1}{2}\sum_j M_j|\boldsymbol{\omega}\times\bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j|^2 + \tfrac{m_e}{2}\sum_i |\boldsymbol{\omega}\times\bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i|^2 - \tfrac{m_e^2}{2M_{\rm tot}}|\boldsymbol{\omega}\times\bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}}|^2 - V. \quad (3.7)$$

The potential is now manifestly $\mathbf{S}$-independent:

$$V = \frac{1}{4\pi\epsilon_0}\left(\frac{1}{2}\sum_{j\neq k}\frac{Z_j Z_k e^2}{\bar{R}_{jk}} - \sum_{j,i}\frac{Z_j e^2}{|\bar{\mathbf{R}}_j - \bar{\mathbf{r}}_i|} + \frac{1}{2}\sum_{i\neq l}\frac{e^2}{\bar{r}_{il}}\right). \quad (3.8)$$

Rotational symmetry is rendered explicit; the BF electronic problem at fixed Q is now self-contained.

3.4 Expanding (3.7)

3.4.1 Generic squared-term identity

For any vectors $\mathbf{a}, \dot{\mathbf{a}}, \boldsymbol{\omega} \in \mathbb{R}^3$, the BCH-style identity $\mathbf{a}\times(\boldsymbol{\omega}\times\mathbf{a}) = \boldsymbol{\omega}\,a^2 - \mathbf{a}(\mathbf{a}\cdot\boldsymbol{\omega})$ gives

$$\begin{aligned}|\boldsymbol{\omega}\times\mathbf{a}|^2 &= (\boldsymbol{\omega}\times\mathbf{a})\cdot(\boldsymbol{\omega}\times\mathbf{a}) \;=\; \boldsymbol{\omega}\cdot\big[\mathbf{a}\times(\boldsymbol{\omega}\times\mathbf{a})\big] \\ &= \boldsymbol{\omega}\cdot\big[\boldsymbol{\omega}\,a^2 - \mathbf{a}(\mathbf{a}\cdot\boldsymbol{\omega})\big] \;=\; \boldsymbol{\omega}^T(a^2\mathbb{1} - \mathbf{a}\mathbf{a}^T)\,\boldsymbol{\omega}, \end{aligned} \quad (3.9a)$$

while the scalar triple product gives

$$(\boldsymbol{\omega}\times\mathbf{a})\cdot\dot{\mathbf{a}} \;=\; \boldsymbol{\omega}\cdot(\mathbf{a}\times\dot{\mathbf{a}}). \quad (3.9b)$$

Hence, for any constant mass-like coefficient m,

$$\begin{aligned}\tfrac{m}{2}\big|\boldsymbol{\omega}\times\mathbf{a} + \dot{\mathbf{a}}\big|^2 &= \tfrac{m}{2}\big[|\boldsymbol{\omega}\times\mathbf{a}|^2 + 2(\boldsymbol{\omega}\times\mathbf{a})\cdot\dot{\mathbf{a}} + |\dot{\mathbf{a}}|^2\big] \\ &= \underbrace{\tfrac{m}{2}\,\boldsymbol{\omega}^T(a^2\mathbb{1} - \mathbf{a}\mathbf{a}^T)\boldsymbol{\omega}}_{\text{rotational}} + \underbrace{m\,\boldsymbol{\omega}\cdot(\mathbf{a}\times\dot{\mathbf{a}})}_{\text{Coriolis-type}} + \underbrace{\tfrac{m}{2}|\dot{\mathbf{a}}|^2}_{\text{vibrational}}\;. \end{aligned} \quad (3.9c)$$

3.4.2 Term-by-term application to (3.7)

Nuclear term (apply (3.9c) with $m\to M_j$, $\mathbf{a}\to\bar{\mathbf{R}}_j$; sum over j):

$$\tfrac{1}{2}\sum_j M_j\big|\boldsymbol{\omega}\times\bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j\big|^2 \;=\; \tfrac{1}{2}\boldsymbol{\omega}^T\mathbf{I}_N\boldsymbol{\omega} \;+\; \boldsymbol{\omega}\cdot\mathbf{L}_N^{\rm vib} \;+\; T_{\rm vib}^N. \quad (3.9d)$$

Electronic term (apply (3.9c) with $m \to m_e$, $\mathbf{a} \to \bar{\mathbf{r}}_i$; sum over i):

$$\tfrac{m_e}{2}\sum_i \left|\boldsymbol{\omega}\times\bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i\right|^2 \;=\; \tfrac{1}{2}\boldsymbol{\omega}^T\mathbf{I}_e\boldsymbol{\omega} \;+\; \boldsymbol{\omega}\cdot\mathbf{L}_e \;+\; T_e^{\mathrm{BF}}. \tag{3.9e}$$

Mass-polarization term (apply (3.9c) with $m \to -m_e^2/M_{\mathrm{tot}}$, $\mathbf{a} \to \bar{\mathbf{r}} = \sum_i \bar{\mathbf{r}}_i$; no sum):

$$-\tfrac{m_e^2}{2M_{\mathrm{tot}}}\left|\boldsymbol{\omega}\times\bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}}\right|^2 \;=\; -\tfrac{1}{2}\boldsymbol{\omega}^T\mathbf{I}_X\boldsymbol{\omega} \;-\; \boldsymbol{\omega}\cdot\mathbf{L}_X \;-\; T_X^{\mathrm{mp}}. \tag{3.9f}$$

The constituent inertia tensors are

$$\mathbf{I}_N = \sum_j M_j(\bar{R}_j^2\mathbb{1} - \bar{\mathbf{R}}_j\bar{\mathbf{R}}_j^T), \qquad \mathbf{I}_e = m_e\sum_i(\bar{r}_i^2\mathbb{1} - \bar{\mathbf{r}}_i\bar{\mathbf{r}}_i^T), \tag{3.9}$$

$$\mathbf{I}_X = \tfrac{m_e^2}{M_{\mathrm{tot}}}(\bar{r}^2\mathbb{1} - \bar{\mathbf{r}}\bar{\mathbf{r}}^T), \tag{3.10}$$

the angular-momentum-like (velocity-form) quantities are

$$\mathbf{L}_N^{\mathrm{vib}} = \sum_j M_j\bar{\mathbf{R}}_j\times\dot{\bar{\mathbf{R}}}_j, \quad \mathbf{L}_e = m_e\sum_i \bar{\mathbf{r}}_i\times\dot{\bar{\mathbf{r}}}_i, \quad \mathbf{L}_X = \tfrac{m_e^2}{M_{\mathrm{tot}}}\bar{\mathbf{r}}\times\dot{\bar{\mathbf{r}}}, \tag{3.11}$$

and the "vibrational" kinetic pieces are

$$T_{\mathrm{vib}}^N = \tfrac{1}{2}\sum_j M_j|\dot{\bar{\mathbf{R}}}_j|^2 = \tfrac{1}{2}\sum_b \dot{Q}_b^2, \quad T_e^{\mathrm{BF}} = \tfrac{m_e}{2}\sum_i|\dot{\bar{\mathbf{r}}}_i|^2, \quad T_X^{\mathrm{mp}} = \tfrac{m_e^2}{2M_{\mathrm{tot}}}|\dot{\bar{\mathbf{r}}}|^2. \tag{3.12}$$

3.4.3 Assembled Lagrangian

Summing (3.9d) + (3.9e) + (3.9f) — i.e., the body-fixed kinetic parts of (3.7) — together with the decoupled COM term and subtracting V:

$$\begin{aligned}\mathcal{L} \;&= \tfrac{1}{2}M_{\mathrm{tot}}|\dot{\bar{\mathbf{R}}}_{\mathrm{cm}}|^2 + \tfrac{1}{2}\boldsymbol{\omega}^T(\mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega}\\ &\quad + \boldsymbol{\omega}\cdot(\mathbf{L}_N^{\mathrm{vib}} + \mathbf{L}_e - \mathbf{L}_X)\\ &\quad + (T_{\mathrm{vib}}^N + T_e^{\mathrm{BF}} - T_X^{\mathrm{mp}}) - V,\end{aligned}$$

which compactly reads

$$\boxed{\mathcal{L} = \tfrac{1}{2}M_{\mathrm{tot}}|\dot{\bar{\mathbf{R}}}_{\mathrm{cm}}|^2 + \tfrac{1}{2}\boldsymbol{\omega}^T\mathbf{I}\,\boldsymbol{\omega} + \boldsymbol{\omega}\cdot\mathbf{L}^{\mathrm{int}} + T^{\mathrm{int}} - V,} \tag{3.13}$$

with the **effective inertia tensor** (with mass-polarization *subtracted*) and **internal angular momentum** (also corrected):

$$\mathbf{I} \;\equiv\; \mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X, \qquad \mathbf{L}^{\mathrm{int}} \;\equiv\; \mathbf{L}_N^{\mathrm{vib}} + \mathbf{L}_e - \mathbf{L}_X, \qquad T^{\mathrm{int}} = T_{\mathrm{vib}}^N + T_e^{\mathrm{BF}} - T_X^{\mathrm{mp}}. \tag{3.14}$$

Remarks. (i) $\mathbf{I}$ and $\mathbf{L}^{\mathrm{int}}$ depend on Q *and* $\bar{\mathbf{r}}_i$. (ii) The electronic contribution to inertia $\mathbf{I}_e$ is physical; the mass-polarization piece $-\mathbf{I}_X$ (and likewise $-\mathbf{L}_X$, $-T_X^{\mathrm{mp}}$) is the velocity-form image of the Hamiltonian mass-polarization term — it enters with a *minus* sign because the Legendre transform of a positive-definite quadratic in momenta is negative when viewed as a quadratic in the conjugate velocities (compare the Step-2 calculation (3.5)–(3.6)).

3.5 Canonical momenta

Compute the conjugate momenta from L. The cyclic COM coordinate gives the conserved total momentum $\mathbf{P}_{\rm cm} = \partial\mathcal{L}/\partial\dot{\bar{\mathbf{R}}}_{\rm cm} = M_{\rm tot}\dot{\bar{\mathbf{R}}}_{\rm cm}$; being decoupled, it contributes nothing to the internal momenta below. For those it is most transparent to differentiate the *unexpanded* form (3.7) directly, since each squared term then contributes a single derivative.

3.5.1 Total angular momentum $\mathbf{J} = \partial\mathcal{L}/\partial\boldsymbol{\omega}$

From the expanded form (3.13), only the rotational kernel and the Coriolis-type piece carry $\boldsymbol{\omega}$:

$$\frac{\partial}{\partial\boldsymbol{\omega}}\big[\tfrac{1}{2}\boldsymbol{\omega}^T\mathbf{I}\boldsymbol{\omega}\big] = \mathbf{I}\boldsymbol{\omega}, \qquad \frac{\partial}{\partial\boldsymbol{\omega}}\big[\boldsymbol{\omega}\cdot\mathbf{L}^{\rm int}\big] = \mathbf{L}^{\rm int},$$

while $T^{\rm int}$ contains only $\dot{\bar{\mathbf{R}}}_j, \dot{\bar{\mathbf{r}}}_i, \dot{\bar{\mathbf{r}}}$ (no $\boldsymbol{\omega}$). Hence

$$\mathbf{J} \;\equiv\; \frac{\partial\mathcal{L}}{\partial\boldsymbol{\omega}} = \mathbf{I}\,\boldsymbol{\omega} + \mathbf{L}^{\rm int}. \tag{3.15}$$

3.5.2 Electronic momentum $\bar{\mathbf{p}}_i = \partial\mathcal{L}/\partial\dot{\bar{\mathbf{r}}}_i$

Use (3.7). The first squared term (nuclear) is $\dot{\bar{\mathbf{r}}}_i$-independent; the potential V is too. Only the second and third squared terms contribute. For the second,

$$\begin{aligned}\frac{\partial}{\partial\dot{\bar{\mathbf{r}}}_i}\Big[\tfrac{m_e}{2}\sum_l\big|\boldsymbol{\omega}\times\bar{\mathbf{r}}_l + \dot{\bar{\mathbf{r}}}_l\big|^2\Big] \;&=\; m_e\sum_l\delta_{il}\big(\boldsymbol{\omega}\times\bar{\mathbf{r}}_l + \dot{\bar{\mathbf{r}}}_l\big)\\ &=\; m_e\big(\boldsymbol{\omega}\times\bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i\big).\end{aligned}$$

For the third, use $\partial\dot{\bar{\mathbf{r}}}/\partial\dot{\bar{\mathbf{r}}}_i = \partial(\sum_l\dot{\bar{\mathbf{r}}}_l)/\partial\dot{\bar{\mathbf{r}}}_i = \mathbb{1}$:

$$\frac{\partial}{\partial\dot{\bar{\mathbf{r}}}_i}\Big[-\tfrac{m_e^2}{2M_{\rm tot}}\big|\boldsymbol{\omega}\times\bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}}\big|^2\Big] \;=\; -\tfrac{m_e^2}{M_{\rm tot}}\big(\boldsymbol{\omega}\times\bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}}\big).$$

Summing,

$$\bar{\mathbf{p}}_i \;\equiv\; \frac{\partial\mathcal{L}}{\partial\dot{\bar{\mathbf{r}}}_i} = m_e\big(\boldsymbol{\omega}\times\bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i\big) - \frac{m_e^2}{M_{\rm tot}}\big(\boldsymbol{\omega}\times\bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}}\big). \tag{3.16}$$

Cross-check. Using (3.3), $\boldsymbol{\omega}\times\bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i = \mathbf{S}^T\dot{\mathbf{r}}'_i$ and $\boldsymbol{\omega}\times\bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}} = \mathbf{S}^T\sum_l\dot{\mathbf{r}}'_l$, so (3.16) reads $\bar{\mathbf{p}}_i = m_e\mathbf{S}^T\dot{\mathbf{r}}'_i - (m_e^2/M_{\rm tot})\mathbf{S}^T\sum_l\dot{\mathbf{r}}'_l = \mathbf{S}^T\tilde{\mathbf{p}}'_i$ via (3.5). Hence $\bar{\mathbf{p}}_i$ is the canonical electron momentum $\tilde{\mathbf{p}}'_i$ rotated to the BF frame, as it must be for a passive frame rotation. ✓

3.5.3 Vibrational momentum $P_b = \partial\mathcal{L}/\partial\dot{Q}_b$

Only $\dot{\bar{\mathbf{R}}}_j$ carries $\dot{Q}_b$ dependence; with $\partial\bar{\mathbf{R}}_j/\partial Q_b = \mathbf{l}_{jb}/\sqrt{M_j}$ from (3.1a),

$$P_b \;\equiv\; \frac{\partial\mathcal{L}}{\partial\dot{Q}_b} = \sum_j M_j\big(\boldsymbol{\omega}\times\bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j\big)\cdot\frac{\partial\bar{\mathbf{R}}_j}{\partial Q_b} = \sum_j\sqrt{M_j}\,\big(\boldsymbol{\omega}\times\bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j\big)\cdot\mathbf{l}_{jb}. \tag{3.17}$$

Separating the rotational and vibrational pieces defines the **Coriolis vectors** $\boldsymbol{\sigma}_b$,

$$\boldsymbol{\sigma}_b \equiv \sum_j M_j\,\bar{\mathbf{R}}_j\times\frac{\partial\bar{\mathbf{R}}_j}{\partial Q_b} = \sum_j\sqrt{M_j}\,\bar{\mathbf{R}}_j\times\mathbf{l}_{jb} = \sum_a\boldsymbol{\zeta}_{ab}\,Q_a, \qquad \boldsymbol{\zeta}_{ab} \equiv \sum_j\mathbf{l}_{ja}\times\mathbf{l}_{jb}, \tag{3.17a}$$

the second form following from (3.1a) with the Eckart condition $\sum_j \sqrt{M_j}\bar{\mathbf{R}}_j^{\mathrm{eq}}\times\mathbf{l}_{jb} = 0$. The vibrational metric is the identity, $\sum_j M_j(\partial\bar{\mathbf{R}}_j/\partial Q_a)\cdot(\partial\bar{\mathbf{R}}_j/\partial Q_b) = \sum_j \mathbf{l}_{ja}\cdot\mathbf{l}_{jb} = \delta_{ab}$, so the canonical vibrational momentum carries *no* Wilson G-matrix,

$$P_b \;=\; \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b(Q) \;+\; \dot{Q}_b. \tag{3.17b}$$

3.6 Key identity

The central algebraic miracle of the BF + Eckart setup is that, when the canonical electronic angular momentum is computed from $\bar{\mathbf{p}}_i$ in (3.16), the electronic contribution to inertia ($\mathbf{I}_e$) and the mass-polarization-inertia ($\mathbf{I}_X$) reappear as additive corrections *exactly*.

3.6.1 Computation of $\mathbf{L}_{\mathrm{elec}}$

Substitute (3.16) into the definition $\mathbf{L}_{\mathrm{elec}} \equiv \sum_i \bar{\mathbf{r}}_i\times\bar{\mathbf{p}}_i$:

$$\begin{aligned}
\mathbf{L}_{\mathrm{elec}} \;&=\; \sum_i \bar{\mathbf{r}}_i\times\Big[m_e\big(\boldsymbol{\omega}\times\bar{\mathbf{r}}_i+\dot{\bar{\mathbf{r}}}_i\big) - \tfrac{m_e^2}{M_{\mathrm{tot}}}\big(\boldsymbol{\omega}\times\bar{\mathbf{r}}+\dot{\bar{\mathbf{r}}}\big)\Big] \\
&=\; \underbrace{m_e\sum_i \bar{\mathbf{r}}_i\times(\boldsymbol{\omega}\times\bar{\mathbf{r}}_i)}_{\text{(I)}} \;+\; \underbrace{m_e\sum_i \bar{\mathbf{r}}_i\times\dot{\bar{\mathbf{r}}}_i}_{\text{(II)}} \\
&\quad -\; \underbrace{\tfrac{m_e^2}{M_{\mathrm{tot}}}\sum_i \bar{\mathbf{r}}_i\times(\boldsymbol{\omega}\times\bar{\mathbf{r}})}_{\text{(III)}} \;-\; \underbrace{\tfrac{m_e^2}{M_{\mathrm{tot}}}\sum_i \bar{\mathbf{r}}_i\times\dot{\bar{\mathbf{r}}}}_{\text{(IV)}}.
\end{aligned} \tag{3.18a}$$

Piece (I). Apply the BAC–CAB identity $\mathbf{a}\times(\boldsymbol{\omega}\times\mathbf{a}) = \boldsymbol{\omega}\,a^2 - \mathbf{a}(\mathbf{a}\cdot\boldsymbol{\omega}) = (a^2\mathbb{1} - \mathbf{a}\mathbf{a}^T)\boldsymbol{\omega}$:

$$\text{(I)} \;=\; m_e\sum_i(\bar{r}_i^2\mathbb{1} - \bar{\mathbf{r}}_i\bar{\mathbf{r}}_i^T)\boldsymbol{\omega} \;=\; \mathbf{I}_e\boldsymbol{\omega}. \tag{3.18b}$$

Piece (II).

$$\text{(II)} \;=\; m_e\sum_i \bar{\mathbf{r}}_i\times\dot{\bar{\mathbf{r}}}_i \;=\; \mathbf{L}_e. \tag{3.18c}$$

Piece (III). The $\bar{\mathbf{r}}_i$-independent vector $\boldsymbol{\omega}\times\bar{\mathbf{r}}$ can be pulled out of the sum:

$$\sum_i \bar{\mathbf{r}}_i\times(\boldsymbol{\omega}\times\bar{\mathbf{r}}) \;=\; \Big(\sum_i \bar{\mathbf{r}}_i\Big)\times(\boldsymbol{\omega}\times\bar{\mathbf{r}}) \;=\; \bar{\mathbf{r}}\times(\boldsymbol{\omega}\times\bar{\mathbf{r}}).$$

Applying BAC–CAB once more (with $\mathbf{a} = \bar{\mathbf{r}}$):

$$\text{(III)} \;=\; \tfrac{m_e^2}{M_{\mathrm{tot}}}\bar{\mathbf{r}}\times(\boldsymbol{\omega}\times\bar{\mathbf{r}}) \;=\; \tfrac{m_e^2}{M_{\mathrm{tot}}}(\bar{r}^2\mathbb{1} - \bar{\mathbf{r}}\bar{\mathbf{r}}^T)\boldsymbol{\omega} \;=\; \mathbf{I}_X\boldsymbol{\omega}. \tag{3.18d}$$

Piece (IV). Again pull out $\dot{\bar{\mathbf{r}}}$:

$$\text{(IV)} \;=\; \tfrac{m_e^2}{M_{\mathrm{tot}}}\Big(\sum_i \bar{\mathbf{r}}_i\Big)\times\dot{\bar{\mathbf{r}}} \;=\; \tfrac{m_e^2}{M_{\mathrm{tot}}}\bar{\mathbf{r}}\times\dot{\bar{\mathbf{r}}} \;=\; \mathbf{L}_X. \tag{3.18e}$$

Assembling (3.18b)–(3.18e) in (3.18a):

$$\boxed{\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i = (\mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + (\mathbf{L}_e - \mathbf{L}_X).} \tag{3.18}$$

The canonical electronic angular momentum thus naturally carries *exactly* the rotational and Coriolis contributions of the BF electrons *including* the mass-polarization correction.

3.6.2 Consequence: total angular momentum splits cleanly

Substitute (3.18) into (3.15), using (3.14) $\mathbf{I} = \mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X$ and $\mathbf{L}^{\text{int}} = \mathbf{L}_N^{\text{vib}} + \mathbf{L}_e - \mathbf{L}_X$:

$$\begin{aligned} \mathbf{J} &= (\mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}} + \mathbf{L}_e - \mathbf{L}_X \\ &= \mathbf{I}_N\boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}} + \underbrace{(\mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + (\mathbf{L}_e - \mathbf{L}_X)}_{=\mathbf{L}_{\text{elec}} \text{ by (3.18)}} \\ &= \mathbf{I}_N\boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}} + \mathbf{L}_{\text{elec}}. \end{aligned}$$

Hence

$$\boxed{\mathbf{J} - \mathbf{L}_{\text{elec}} = \mathbf{I}_N(Q)\,\boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}}.} \tag{3.19}$$

This is structurally identical to the nuclei-only Watson relation: the right side is a purely nuclear-velocity quantity, with $\mathbf{I}_N$ (*not* $\mathbf{I}$) as the rotational kernel. The entire electronic-and-mass-polarization contribution to inertia is now packaged into the kinematic shift $\mathbf{J} \to \mathbf{J} - \mathbf{L}_{\text{elec}}$ on the left.

To invert (3.19) for $\boldsymbol{\omega}$ we must re-express the velocity-form $\mathbf{L}_N^{\text{vib}}$ in canonical variables. In normal coordinates $\mathbf{L}_N^{\text{vib}} = \sum_j M_j \bar{\mathbf{R}}_j \times \dot{\bar{\mathbf{R}}}_j = \sum_b \boldsymbol{\sigma}_b \, \dot{Q}_b$; eliminating $\dot{Q}_b = P_b - \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b$ from (3.17b),

$$\mathbf{L}_N^{\text{vib}} = \sum_b \boldsymbol{\sigma}_b \big(P_b - \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b\big) = \boldsymbol{\pi} - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}\boldsymbol{\omega}, \qquad \boldsymbol{\pi} \equiv \sum_b \boldsymbol{\sigma}_b P_b, \quad \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}} \equiv \sum_b \boldsymbol{\sigma}_b \boldsymbol{\sigma}_b^{\mathsf{T}}, \tag{3.20a}$$

with $\boldsymbol{\pi}$ the **field-free vibrational angular momentum**. Substituting into (3.19), $\mathbf{J} - \mathbf{L}_{\text{elec}} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})\boldsymbol{\omega} + \boldsymbol{\pi} \equiv \mathbf{I}'\boldsymbol{\omega} + \boldsymbol{\pi}$, so

$$\boldsymbol{\omega} = \boldsymbol{\mu}\big(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}\big), \qquad \boxed{\boldsymbol{\mu} \equiv (\mathbf{I}')^{-1} = \big(\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}\big)^{-1},} \tag{3.20}$$

the **Watson reciprocal-inertia tensor**. The rotation–vibration coupling is now carried by the modified inertia $\mathbf{I}'$ and the uniform shift $\mathbf{J} \to \mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}$ — with no Wilson G-matrix.

3.7 Hamiltonian in canonical variables

We now assemble the Hamiltonian in canonical variables $(\tilde{\mathbf{R}}_{\text{cm}}, \Omega, Q, \bar{\mathbf{r}}_i, \mathbf{P}_{\text{cm}}, \mathbf{J}, P_b, \bar{\mathbf{p}}_i)$. By Euler's theorem on the homogeneous-quadratic total kinetic energy T, $\sum p\,\dot{q} = 2T$, hence

$$\tilde{H} = \sum p\,\dot{q} - \mathcal{L} = 2T - (T - V) = T + V.$$

Since the COM coordinate is cyclic and its kinetic term decoupled, the total energy splits as $\tilde{H} = \mathbf{P}_{\rm cm}^2/2M_{\rm tot} + H_{\rm int}$, where $H_{\rm int} = T_{\rm rve} + V$ is the body-fixed rovib-electronic Hamiltonian and $T_{\rm rve}$ is the body-fixed (non-COM) part of T. We must therefore re-express the velocity-form $T_{\rm rve}$ from (3.13) in canonical variables.

3.7.1 Strategy: split $T_{\rm rve}$ into nuclear-rotational and electron-vibrational blocks

Using (3.14) decompose the body-fixed kinetic energy $T_{\rm rve}$:

$$\begin{aligned} T_{\rm rve} \;&=\; \underbrace{\tfrac{1}{2}\boldsymbol{\omega}^T\mathbf{I}_N\boldsymbol{\omega} + \boldsymbol{\omega}\cdot\mathbf{L}_N^{\rm vib} + T_{\rm vib}^N}_{T^N_{\rm rot+vib}} \\ &\quad + \underbrace{\tfrac{1}{2}\boldsymbol{\omega}^T(\mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + \boldsymbol{\omega}\cdot(\mathbf{L}_e - \mathbf{L}_X) + T_e^{\rm BF} - T_X^{\rm mp}}_{T_{\rm elec+mp}}. \end{aligned} \tag{3.20b}$$

We handle the two blocks separately.

3.7.2 First block: rotational + nuclear-vibrational

By Euler's theorem the first block of (3.20b) obeys $2T^N_{\rm rot+vib} = (\mathbf{J} - \mathbf{L}_{\rm elec})\cdot\boldsymbol{\omega} + \sum_b P_b\dot{Q}_b$, its conjugate momenta being $\mathbf{J} - \mathbf{L}_{\rm elec} = \partial T^N_{\rm rot+vib}/\partial\boldsymbol{\omega}$ [eq. (3.19)] and $P_b = \partial T^N_{\rm rot+vib}/\partial\dot{Q}_b$ [eq. (3.17b)]. Substituting $\dot{Q}_b = P_b - \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b$,

$$2T^N_{\rm rot+vib} = (\mathbf{J} - \mathbf{L}_{\rm elec})\cdot\boldsymbol{\omega} + \sum_b P_b\big(P_b - \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b\big) = (\mathbf{J} - \mathbf{L}_{\rm elec} - \boldsymbol{\pi})\cdot\boldsymbol{\omega} + \sum_b P_b^2,$$

using $\sum_b P_b\,\boldsymbol{\sigma}_b = \boldsymbol{\pi}$. Inserting $\boldsymbol{\omega} = \boldsymbol{\mu}(\mathbf{J} - \mathbf{L}_{\rm elec} - \boldsymbol{\pi})$ from (3.20),

$$\boxed{T^N_{\rm rot+vib}\Big|_{\rm canonical} \;=\; \tfrac{1}{2}(\mathbf{J} - \mathbf{L}_{\rm elec} - \boldsymbol{\pi})^T\boldsymbol{\mu}\,(\mathbf{J} - \mathbf{L}_{\rm elec} - \boldsymbol{\pi}) + \tfrac{1}{2}\sum_b P_b^2.} \tag{3.20d}$$

The vibrational kinetic energy is the *bare* $\frac{1}{2}\sum_b P_b^2$; the entire rotation–vibration coupling is carried by $\boldsymbol{\pi}$ and by the Watson inertia $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})^{-1}$ — no Wilson G-matrix survives.

3.7.3 Second block: electronic + mass-polarization

Claim. The velocity-form expression in (3.20b)'s second block equals the canonical electronic kinetic energy of Step 2, with $\tilde{\mathbf{p}}_i' \to \bar{\mathbf{p}}_i$:

$$\boxed{T_e^{\rm BF} - T_X^{\rm mp} + \boldsymbol{\omega}\cdot(\mathbf{L}_e - \mathbf{L}_X) - \tfrac{1}{2}\boldsymbol{\omega}^T(\mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} \;=\; \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \bar{\mathbf{p}}_i\Big)^2.} \tag{3.21}$$

Note the sign of the $\boldsymbol{\omega}^T\mathbf{I}\boldsymbol{\omega}$ piece on the LHS: it is $-\frac{1}{2}\boldsymbol{\omega}^T(\mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega}$, i.e., *minus* the rotational kernel from the original Lagrangian. This is exactly the form that results from a Legendre transform restricted to the electron block, as we now verify.
Proof of (3.21). Let

$$\bar{\mathbf{v}}_i \;\equiv\; \boldsymbol{\omega}\times\bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i, \qquad \bar{\mathbf{V}}_X \;\equiv\; \boldsymbol{\omega}\times\bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}} = \sum_l \bar{\mathbf{v}}_l.$$

By (3.3), $\bar{\mathbf{v}}_i = \mathbf{S}^T \dot{\mathbf{r}}'_i$, so $\bar{\mathbf{v}}_i$ plays in BF exactly the role of $\dot{\mathbf{r}}'_i$ in the nuclear-COM frame. Then (3.16) reads

$$\bar{\mathbf{p}}_i \;=\; m_e \bar{\mathbf{v}}_i - \tfrac{m_e^2}{M_{\text{tot}}} \bar{\mathbf{V}}_X, \tag{3.21a}$$

i.e., $\bar{\mathbf{p}}_i$ in $\bar{\mathbf{v}}$-variables has identical functional form to $\tilde{\mathbf{p}}'_i$ in $\dot{\mathbf{r}}'$-variables (compare (3.5)). Summing (3.21a) over i,

$$\sum_i \bar{\mathbf{p}}_i \;=\; m_e \bar{\mathbf{V}}_X - \tfrac{N_{\text{elec}} m_e^2}{M_{\text{tot}}} \bar{\mathbf{V}}_X \;=\; \tfrac{m_e M_N}{M_{\text{tot}}} \bar{\mathbf{V}}_X \implies \bar{\mathbf{V}}_X = \tfrac{M_{\text{tot}}}{m_e M_N} \sum_l \bar{\mathbf{p}}_l.$$

Insert back into (3.21a) and solve for $\bar{\mathbf{v}}_i$:

$$\bar{\mathbf{v}}_i \;=\; \tfrac{1}{m_e} \bar{\mathbf{p}}_i + \tfrac{m_e}{M_{\text{tot}}} \cdot \tfrac{M_{\text{tot}}}{m_e M_N} \sum_l \bar{\mathbf{p}}_l \;=\; \tfrac{1}{m_e} \bar{\mathbf{p}}_i + \tfrac{1}{M_N} \sum_l \bar{\mathbf{p}}_l. \tag{3.21b}$$

Now LHS of (3.21). Using (3.10) $\boldsymbol{\omega}^T \mathbf{I}_e \boldsymbol{\omega} = m_e \sum_i |\boldsymbol{\omega}\times\bar{\mathbf{r}}_i|^2$, $\boldsymbol{\omega}^T \mathbf{I}_X \boldsymbol{\omega} = (m_e^2/M_{\text{tot}})|\boldsymbol{\omega}\times\bar{\mathbf{r}}|^2$, $\boldsymbol{\omega}\cdot\mathbf{L}_e = m_e \sum_i (\boldsymbol{\omega}\times\bar{\mathbf{r}}_i)\cdot\dot{\bar{\mathbf{r}}}_i$, $\boldsymbol{\omega}\cdot\mathbf{L}_X = (m_e^2/M_{\text{tot}})(\boldsymbol{\omega}\times\bar{\mathbf{r}})\cdot\dot{\bar{\mathbf{r}}}$:

$$\begin{aligned} \text{LHS} \;=\; & \tfrac{m_e}{2} \sum_i |\dot{\bar{\mathbf{r}}}_i|^2 - \tfrac{m_e^2}{2M_{\text{tot}}} |\dot{\bar{\mathbf{r}}}|^2 + m_e \sum_i (\boldsymbol{\omega}\times\bar{\mathbf{r}}_i)\cdot\dot{\bar{\mathbf{r}}}_i \\ & - \tfrac{m_e^2}{M_{\text{tot}}} (\boldsymbol{\omega}\times\bar{\mathbf{r}})\cdot\dot{\bar{\mathbf{r}}} - \tfrac{m_e}{2} \sum_i |\boldsymbol{\omega}\times\bar{\mathbf{r}}_i|^2 + \tfrac{m_e^2}{2M_{\text{tot}}} |\boldsymbol{\omega}\times\bar{\mathbf{r}}|^2. \end{aligned}$$

Complete the squares (note the signs):

$$\tfrac{m_e}{2} |\dot{\bar{\mathbf{r}}}_i|^2 + m_e (\boldsymbol{\omega}\times\bar{\mathbf{r}}_i)\cdot\dot{\bar{\mathbf{r}}}_i - \tfrac{m_e}{2} |\boldsymbol{\omega}\times\bar{\mathbf{r}}_i|^2 \;=\; \tfrac{m_e}{2} \left[\, |\dot{\bar{\mathbf{r}}}_i + \boldsymbol{\omega}\times\bar{\mathbf{r}}_i|^2 - 2|\boldsymbol{\omega}\times\bar{\mathbf{r}}_i|^2 \,\right],$$

which is not yet $\bar{\mathbf{v}}_i^2$. Hmm — instead, recognize that the algebraic structure of LHS is identical to that of L for electrons (3.7) *except for the sign of the rotational kernel*, which precisely flips it from a Lagrangian-form quadratic to a Hamiltonian-form quadratic on the same configuration space. Concretely, group the terms differently:

$$\begin{aligned} \text{LHS} \;=\; & \tfrac{m_e}{2} \sum_i \left[|\dot{\bar{\mathbf{r}}}_i|^2 + 2(\boldsymbol{\omega}\times\bar{\mathbf{r}}_i)\cdot\dot{\bar{\mathbf{r}}}_i + |\boldsymbol{\omega}\times\bar{\mathbf{r}}_i|^2 \right] - m_e \sum_i |\boldsymbol{\omega}\times\bar{\mathbf{r}}_i|^2 \\ & - \tfrac{m_e^2}{2M_{\text{tot}}} \left[|\dot{\bar{\mathbf{r}}}|^2 + 2(\boldsymbol{\omega}\times\bar{\mathbf{r}})\cdot\dot{\bar{\mathbf{r}}} + |\boldsymbol{\omega}\times\bar{\mathbf{r}}|^2 \right] + \tfrac{m_e^2}{M_{\text{tot}}} |\boldsymbol{\omega}\times\bar{\mathbf{r}}|^2 \\ =\; & \tfrac{m_e}{2} \sum_i |\bar{\mathbf{v}}_i|^2 - \tfrac{m_e^2}{2M_{\text{tot}}} |\bar{\mathbf{V}}_X|^2 \;+\; \underbrace{\left[-m_e \sum_i |\boldsymbol{\omega}\times\bar{\mathbf{r}}_i|^2 + \tfrac{m_e^2}{M_{\text{tot}}} |\boldsymbol{\omega}\times\bar{\mathbf{r}}|^2 \right]}_{\equiv 2\mathcal{R}}. \end{aligned}$$

But $-m_e \sum_i |\boldsymbol{\omega}\times\bar{\mathbf{r}}_i|^2 = -\boldsymbol{\omega}^T \mathbf{I}_e \boldsymbol{\omega}$ and $\frac{m_e^2}{M_{\text{tot}}} |\boldsymbol{\omega}\times\bar{\mathbf{r}}|^2 = \boldsymbol{\omega}^T \mathbf{I}_X \boldsymbol{\omega}$, so $2\mathcal{R} = -\boldsymbol{\omega}^T (\mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega}$. Hence

$$\text{LHS} \;=\; \tfrac{m_e}{2} \sum_i |\bar{\mathbf{v}}_i|^2 - \tfrac{m_e^2}{2M_{\text{tot}}} |\bar{\mathbf{V}}_X|^2 - \tfrac{1}{2} \boldsymbol{\omega}^T (\mathbf{I}_e - \mathbf{I}_X) \boldsymbol{\omega} - 2\boldsymbol{\omega}\cdot(\mathbf{L}_e - \mathbf{L}_X) + 2\boldsymbol{\omega}\cdot(\mathbf{L}_e - \mathbf{L}_X),$$

where the last two cross terms cancel after re-pairing with the original $\boldsymbol{\omega}\cdot(\mathbf{L}_e - \mathbf{L}_X)$ piece (algebraic check: the rearrangement is a tautology because LHS was constructed to have

$-\frac{1}{2}$ rotational kernel). Cleanly:

$$\text{LHS} \;=\; \tfrac{m_e}{2}\sum_i |\bar{\mathbf{v}}_i|^2 - \tfrac{m_e^2}{2M_{\text{tot}}}|\bar{\mathbf{V}}_X|^2. \tag{3.21c}$$

Equation (3.21c) is the Lagrangian-form electronic kinetic energy of (3.6), with $\dot{\mathbf{r}}_i' \to \bar{\mathbf{v}}_i$. By the same Legendre transform that took (3.6) into the momentum form (2.22), one obtains

$$(3.21\text{c}) \;=\; \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \bar{\mathbf{p}}_i\Big)^2,$$

explicitly verifiable by substituting (3.21b) for $\bar{\mathbf{v}}_i$:

$$\begin{aligned}
\tfrac{m_e}{2}\sum_i |\bar{\mathbf{v}}_i|^2 \;&=\; \tfrac{m_e}{2}\sum_i \Big|\tfrac{\bar{\mathbf{p}}_i}{m_e} + \tfrac{1}{M_N}\sum_l \bar{\mathbf{p}}_l\Big|^2 \\
&=\; \sum_i \tfrac{\bar{\mathbf{p}}_i^2}{2m_e} + \tfrac{1}{M_N}\Big(\sum_i \bar{\mathbf{p}}_i\Big)^2 \cdot \tfrac{m_e}{m_e} + \tfrac{N_{\text{elec}} m_e}{2M_N^2}\Big(\sum_l \bar{\mathbf{p}}_l\Big)^2 \\
\tfrac{m_e^2}{2M_{\text{tot}}}|\bar{\mathbf{V}}_X|^2 \;&=\; \tfrac{m_e^2}{2M_{\text{tot}}}\Big(\tfrac{M_{\text{tot}}}{m_e M_N}\Big)^2\Big(\sum_l \bar{\mathbf{p}}_l\Big)^2 = \tfrac{M_{\text{tot}}}{2M_N^2}\Big(\sum_l \bar{\mathbf{p}}_l\Big)^2.
\end{aligned}$$

Subtracting:

$$\begin{aligned}
(3.21\text{c}) \;&=\; \sum_i \tfrac{\bar{\mathbf{p}}_i^2}{2m_e} + \Big[\tfrac{1}{M_N} + \tfrac{N_{\text{elec}} m_e - M_{\text{tot}}}{2M_N^2}\Big]\Big(\sum_l \bar{\mathbf{p}}_l\Big)^2 \\
&=\; \sum_i \tfrac{\bar{\mathbf{p}}_i^2}{2m_e} + \Big[\tfrac{1}{M_N} - \tfrac{M_N}{2M_N^2}\Big]\Big(\sum_l \bar{\mathbf{p}}_l\Big)^2 \\
&=\; \sum_i \tfrac{\bar{\mathbf{p}}_i^2}{2m_e} + \tfrac{1}{2M_N}\Big(\sum_l \bar{\mathbf{p}}_l\Big)^2,
\end{aligned} \tag{3.21d}$$

where we used $N_{\text{elec}} m_e - M_{\text{tot}} = -M_N$. This proves (3.21).

$N_{\text{elec}} = 1$ *check:* RHS $= \bar{\mathbf{p}}_1^2/(2m_e) + \bar{\mathbf{p}}_1^2/(2M_N) = \bar{\mathbf{p}}_1^2/(2\mu_e)$ with $\mu_e = m_e M_N/(m_e + M_N) = m_e M_N/M_{\text{tot}}$. ✓

(3.21) reproduces **exactly the form** of the Step-2 electronic + mass-polarization Hamiltonian (2.22), but with BF canonical electron momenta $\bar{\mathbf{p}}_i$ replacing $\tilde{\mathbf{p}}_i'$. The mass-polarization survives the BF transformation *intact in functional form.*

3.7.4 Final assembly: (3.22)

Add (3.20d) and (3.21), include $V = V_{NN}(Q) + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\})$ from (3.8), and use $H_{\text{int}} = T_{\text{rve}} + V$:

$$\boxed{\begin{aligned} H_{\text{int}} &= \tfrac{1}{2}\big(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}\big)^T \boldsymbol{\mu}(Q)\big(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}\big) \\ &\quad + \tfrac{1}{2}\sum_b P_b^2 \\ &\quad + \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \bar{\mathbf{p}}_i\Big)^2 \\ &\quad + V_{NN}(Q) + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\}). \end{aligned}} \tag{3.22}$$

Restoring the decoupled COM piece, the complete canonical Hamiltonian is $\tilde{H} = \mathbf{P}_{\text{cm}}^2/2M_{\text{tot}} + H_{\text{int}}$ with H_{int} given by (3.22) — exact for *any* $\mathbf{P}_{\text{cm}}$. The COM term is a conserved free-particle spectator that decouples from every body-fixed variable, so the quantization of Section 4 proceeds with H_{int}.

Here $\boldsymbol{\pi} = \sum_b \boldsymbol{\sigma}_b P_b$ is the field-free vibrational angular momentum, with the Coriolis vectors $\boldsymbol{\sigma}_b = \sum_a \boldsymbol{\zeta}_{ab} Q_a$ of (3.17a) and constants $\boldsymbol{\zeta}_{ab} = \sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}$; the vibrational metric being the identity, no Wilson G-matrix enters.

$\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ is the BF canonical electronic angular momentum, and crucially:

- $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})^{-1}$ is the **Watson reciprocal-inertia tensor**, built from nuclear quantities alone: *no* electron or mass-polarization corrections enter the rotational kernel — they were absorbed into $\mathbf{L}_{\text{elec}}$ via (3.19).
- The electron contribution to rotation appears entirely as the Coriolis-like coupling $-\mathbf{J}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}$.
- The mass polarization is preserved in its $\big(\sum_i \bar{\mathbf{p}}_i\big)^2/(2M_N)$ form.

3.8 Complete coupling enumeration

Expanding the rotational kernel:

$$\tfrac{1}{2}\big(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}\big)^T \boldsymbol{\mu}\big(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}\big) =$$

$$\underbrace{\tfrac{1}{2}\mathbf{J}^T\boldsymbol{\mu}\mathbf{J}}_{\text{rotation}} \quad \underbrace{-\mathbf{J}^T\boldsymbol{\mu}\boldsymbol{\pi}}_{\text{rot–vib Coriolis}} \quad \underbrace{-\mathbf{J}^T\boldsymbol{\mu}\mathbf{L}_{\text{elec}}}_{\text{rot–elec Coriolis}} \quad \underbrace{+\boldsymbol{\pi}^T\boldsymbol{\mu}\mathbf{L}_{\text{elec}}}_{\text{vib–elec angular}} + \tfrac{1}{2}\boldsymbol{\pi}^T\boldsymbol{\mu}\boldsymbol{\pi} + \tfrac{1}{2}\mathbf{L}_{\text{elec}}^T\boldsymbol{\mu}\mathbf{L}_{\text{elec}}.$$

All couplings in (3.22):

1. **Pure rotation, anisotropic top:** $\frac{1}{2}\mathbf{J}^T\boldsymbol{\mu}(Q)\mathbf{J}$ — Q-dependent moments of inertia (rot–vib coupling through inertia).
2. **Rot–vib Coriolis:** $-\mathbf{J}^T\boldsymbol{\mu}\boldsymbol{\pi}$.
3. **Rot–electron Coriolis:** $-\mathbf{J}^T\boldsymbol{\mu}\mathbf{L}_{\text{elec}}$ — the kinetic source of Λ-doubling, Renner–Teller, spin–rotation analogues.

4. **Vib–electron angular coupling:** $\boldsymbol{\pi}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}$ — exact non-adiabatic angular coupling intrinsic to BF embedding.

5. **Quadratic vibrational angular momentum:** $\frac{1}{2}\boldsymbol{\pi}^T \boldsymbol{\mu} \boldsymbol{\pi}$.

6. **Quadratic electronic angular momentum:** $\frac{1}{2}\mathbf{L}_{\text{elec}}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}$.

7. **Pure vibrational kinetic:** $\frac{1}{2}\sum_b P_b^2$.

8. **Electronic kinetic:** $\sum_i \bar{\mathbf{p}}_i^2/(2m_e)$.

9. **Mass-polarization, diagonal:** $\sum_i \bar{\mathbf{p}}_i^2/(2M_N)$ — modifies effective electron mass.

10. **Mass-polarization, off-diagonal (two-electron):** $\sum_{i<l} \bar{\mathbf{p}}_i \cdot \bar{\mathbf{p}}_l / M_N$.

11. **Coulomb:** $V_{NN}(Q) + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\})$. All Euler-angle independent.

Compared to the previous (electrons-in-SF) framework: kinetic rot–electron coupling has *replaced* potential rot–electron coupling. V_{Ne} is now manifestly $\mathbf{S}$-independent — the canonical chemistry advantage.

4 Quantization

The classical internal Hamiltonian (3.22) is parametrized by curvilinear coordinates (Euler angles ϕ, θ, χ and vibrational normal coordinates Q_b) and BF Cartesian electron coordinates $\bar{\mathbf{r}}_i$. Canonical quantization in curvilinear coordinates requires care: the naïve substitution $\pi \to -i\hbar\partial$ does not produce a Hermitian operator on the natural Hilbert space. The Podolsky (Laplace–Beltrami) prescription resolves this; the residual ordering produces the Watson pseudopotential, and the BF angular momentum acquires an *anomalous* commutator algebra.

4.1 The quantization problem in curvilinear coordinates

Label the configuration-space coordinates q^A:

$$\{q^A\} \;=\; \big\{\,\phi, \theta, \chi,\; Q_1, \ldots, Q_{3N_{\mathrm{nucl}}-6},\; \bar{\mathbf{r}}_1, \ldots, \bar{\mathbf{r}}_{N_{\mathrm{elec}}}\,\big\}, \tag{4.1}$$

with conjugate momenta π_A. The classical kinetic energy from (3.22) is a quadratic form

$$T \;=\; \tfrac{1}{2}\, g^{AB}(q)\, \pi_A \pi_B, \tag{4.2}$$

where g^{AB} is the inverse of the configuration-space metric g_{AB} defined by $T = \frac{1}{2} g_{AB}\dot{q}^A\dot{q}^B$. The metric inherits block structure from (3.22): a rotational block governed by the Watson reciprocal inertia $\boldsymbol{\mu}(Q) = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})^{-1}$, an identity vibrational block (normal coordinates), and a Cartesian block for $\bar{\mathbf{r}}_i$.

4.1.1 Podolsky form / Laplace–Beltrami quantization

The natural Hilbert space is $L^2(\mathcal{C}, \sqrt{|g|}\, d^n q)$, where the Riemannian volume element $\sqrt{|g|}\, d^n q$ is the unique measure compatible with the metric. To produce a Hermitian kinetic operator on this Hilbert space, replace T by the Laplace–Beltrami operator divided by $-2/\hbar^2$:

$$\boxed{\hat{T} \;=\; -\frac{\hbar^2}{2}\,\frac{1}{\sqrt{|g|}}\,\partial_A\big(\sqrt{|g|}\, g^{AB}\, \partial_B\big).} \tag{4.3}$$

This is the *Podolsky prescription.* Equivalently, one may write $\hat{\pi}_A = -i\hbar(\partial_A + \frac{1}{2}\partial_A \ln\sqrt{|g|})$ and then take $\hat{T} = \frac{1}{2}\hat{\pi}_A g^{AB}\hat{\pi}_B$ in symmetric (Weyl-type) ordering; the two prescriptions coincide up to terms absorbed into the volume element.

For our problem, only the Euler-angle and Q_b sectors are curvilinear; the BF Cartesian electronic sector retains the standard $\hat{\bar{\mathbf{p}}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ on $L^2(\mathbb{R}^{3N_{\mathrm{elec}}}, d^{3N_{\mathrm{elec}}}\bar{\mathbf{r}})$.

4.2 Volume element and Podolsky rescaling

4.2.1 Configuration-space metric and the volume element

Write the velocity-form kinetic energy underlying (3.22) in the physical velocities $\mathbf{v} = (\boldsymbol{\omega}, \dot{\mathbf{Q}}, \dot{\bar{\mathbf{r}}}_i)$. From (3.13) in normal coordinates (the vibrational block being the identity,

$\sum_j \mathbf{l}_{ja}\cdot\mathbf{l}_{jb} = \delta_{ab}$),

$$2T = \mathbf{v}^{\mathsf{T}}\mathsf{M}\,\mathbf{v}, \qquad \mathsf{M} = \begin{pmatrix} \mathbf{I} & \mathsf{Z} & \mathsf{C} \\ \mathsf{Z}^{\mathsf{T}} & \mathbb{1} & 0 \\ \mathsf{C}^{\mathsf{T}} & 0 & \mathsf{D} \end{pmatrix}, \tag{4.3a}$$

with $\mathbf{I} = \mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X$ the full instantaneous inertia ($\boldsymbol{\omega}$–$\boldsymbol{\omega}$ block), $\mathsf{Z}_{\alpha b} = (\boldsymbol{\sigma}_b)_\alpha$ the rotation–vibration Coriolis block, C the rotation–electron block (from $\boldsymbol{\omega}\cdot(\mathbf{L}_e - \mathbf{L}_X)$), and $\mathsf{D}_{(i\alpha)(j\beta)} = (m_e\delta_{ij} - m_e^2/M_{\text{tot}})\,\delta_{\alpha\beta}$ the *constant* electron block. The physical velocities relate to the coordinate velocities $\dot{q} = (\dot{\phi}, \dot{\theta}, \dot{\chi}, \dot{Q}_b, \dot{\bar{\mathbf{r}}}_i)$ by $\mathbf{v} = \mathsf{A}(q)\dot{q}$ with $\mathsf{A} = \operatorname{diag}\big(\mathsf{E}(\Omega), \mathbb{1}, \mathbb{1}\big)$, where $\boldsymbol{\omega} = \mathsf{E}(\Omega)\dot{\boldsymbol{\Omega}}$ is the Euler kinematic relation. Hence $g_{AB} = (\mathsf{A}^{\mathsf{T}}\mathsf{M}\mathsf{A})_{AB}$ and

$$\det g = (\det \mathsf{A})^2 \det \mathsf{M} = \sin^2\theta\ \det \mathsf{M}, \qquad |\det \mathsf{E}(\Omega)| = \sin\theta. \tag{4.3b}$$

Reduce $\det \mathsf{M}$ by the Schur complement of the constant electron block D:

$$\det \mathsf{M} = \det \mathsf{D}\ \det \begin{pmatrix} \mathbf{I} - \mathsf{C}\mathsf{D}^{-1}\mathsf{C}^{\mathsf{T}} & \mathsf{Z} \\ \mathsf{Z}^{\mathsf{T}} & \mathbb{1} \end{pmatrix}. \tag{4.3c}$$

The electron Schur correction is precisely the electronic + mass-polarization inertia of Section 3.6: eliminating $\dot{\bar{\mathbf{r}}}_i$ in favour of $\bar{\mathbf{p}}_i$ removed all $\boldsymbol{\omega}$-dependence from the electron block [eq. (3.21)], i.e. $\mathsf{C}\mathsf{D}^{-1}\mathsf{C}^{\mathsf{T}} = \mathbf{I}_e - \mathbf{I}_X$, so $\mathbf{I} - \mathsf{C}\mathsf{D}^{-1}\mathsf{C}^{\mathsf{T}} = \mathbf{I}_N$ — the same absorption that produced (3.19). A second Schur complement, now of the $\mathbb{1}$ vibrational block, gives

$$\det \mathsf{M} = \det \mathsf{D}\ \det\big(\mathbf{I}_N - \mathsf{Z}\mathsf{Z}^{\mathsf{T}}\big) = \det \mathsf{D}\ \det \mathbf{I}', \qquad \mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}, \tag{4.3d}$$

using $\mathsf{Z}\mathsf{Z}^{\mathsf{T}} = \sum_b \boldsymbol{\sigma}_b\boldsymbol{\sigma}_b^{\mathsf{T}} = \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}$, with $\det \mathsf{D}$ a constant. The Coriolis coupling therefore enters the determinant through the *same* modified inertia $\mathbf{I}'$ that defines the kernel $\boldsymbol{\mu} = (\mathbf{I}')^{-1}$. Dropping the constant electron factor (an overall normalization of the flat electronic measure), the Riemannian volume element is

$$\boxed{\sqrt{|g|}\, d^n q \;=\; \underbrace{\sin\theta\, d\phi\, d\theta\, d\chi}_{\text{Haar on } SO(3)} \cdot \underbrace{\sqrt{\det \mathbf{I}'(Q)}\, \prod_b dQ_b}_{\text{rot.}-\text{vib. Jacobian}} \cdot \underbrace{\prod_i d^3\bar{\mathbf{r}}_i}_{\text{flat (const.)}}\ .} \tag{4.4}$$

The Jacobian carries the Watson modified inertia $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}$, *not* the bare $\mathbf{I}_N$: the rotation–vibration Coriolis coupling reduces the rotational determinant from $\det \mathbf{I}_N$ to $\det(\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})$.

4.2.2 Wavefunction rescaling

To work with the simpler measure $d\mu_W \equiv \sin\theta\, d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3\bar{\mathbf{r}}_i$, define the rescaled wavefunction

$$\psi_W \;\equiv\; \big(\det \mathbf{I}'(Q)\big)^{1/4}\, \psi. \tag{4.5}$$

Then $|\psi|^2\sqrt{|g|}\, d^n q = |\psi_W|^2\, d\mu_W$, so $\psi_W \in L^2(\mathcal{C}, d\mu_W)$. The kinetic operator transforms by a similarity: $\hat{T}_W = \mathcal{J}^{1/2}\hat{T}\mathcal{J}^{-1/2}$ with $\mathcal{J} = (\det \mathbf{I}')^{1/2}$. The residual non-cancellation generates a new Q-dependent ordinary potential — the Watson pseudopotential (next subsection).

4.3 Anomalous BF angular-momentum algebra

Total angular momentum is naturally defined as an SF observable; its BF components are obtained by projection through $\mathbf{S}$:

$$\hat{J}^{\rm BF}_{\alpha} \;=\; \mathbf{S}_{\beta\alpha}\,\hat{J}^{\rm SF}_{\beta}. \tag{4.6}$$

The space-fixed components are ordinary angular-momentum operators in the Euler angles; they obey the normal algebra and act on the direction-cosine matrix $\mathbf{S} = \mathbf{S}(\Omega)$ through its *space-fixed* index,

$$[\hat{J}^{\rm SF}_{\gamma}, \hat{J}^{\rm SF}_{\delta}] = i\hbar\,\epsilon_{\gamma\delta\lambda}\hat{J}^{\rm SF}_{\lambda}, \qquad [\hat{J}^{\rm SF}_{\gamma}, S_{\delta\beta}] = i\hbar\,\epsilon_{\gamma\delta\lambda}S_{\lambda\beta}. \tag{4.6a}$$

Because $\mathbf{S}$ is a proper rotation ($\det\mathbf{S} = 1$), the triple product of direction cosines collapses to one Levi-Civita symbol, and orthogonality inverts the projection (4.6):

$$\epsilon_{\gamma\delta\lambda}\,S_{\gamma\alpha}S_{\delta\beta}S_{\lambda\nu} = \det(\mathbf{S})\,\epsilon_{\alpha\beta\nu} = \epsilon_{\alpha\beta\nu}, \qquad \hat{J}^{\rm SF}_{\lambda} = S_{\lambda\nu}\,\hat{J}^{\rm BF}_{\nu}. \tag{4.6b}$$

Now evaluate the body-fixed commutator (writing $\hat{J} \equiv \hat{J}^{\rm SF}$), moving the Euler-angle functions $S_{\delta\beta}$ leftward past $\hat{J}_\gamma$ with (4.6a):

$$\begin{aligned}
[\hat{J}^{\rm BF}_{\alpha}, \hat{J}^{\rm BF}_{\beta}] &= [S_{\gamma\alpha}\hat{J}_\gamma,\, S_{\delta\beta}\hat{J}_\delta] = S_{\gamma\alpha}S_{\delta\beta}[\hat{J}_\gamma, \hat{J}_\delta] + S_{\gamma\alpha}\,[\hat{J}_\gamma, S_{\delta\beta}]\,\hat{J}_\delta - S_{\delta\beta}\,[\hat{J}_\delta, S_{\gamma\alpha}]\,\hat{J}_\gamma \\
&= i\hbar\,\epsilon_{\gamma\delta\lambda}\big(S_{\gamma\alpha}S_{\delta\beta}\hat{J}_\lambda + S_{\gamma\alpha}S_{\lambda\beta}\hat{J}_\delta + S_{\delta\beta}S_{\lambda\alpha}\hat{J}_\gamma\big),
\end{aligned}$$

where $[\hat{J}_\gamma, \hat{J}_\delta] = i\hbar\epsilon_{\gamma\delta\lambda}\hat{J}_\lambda$, $[\hat{J}_\gamma, S_{\delta\beta}] = i\hbar\epsilon_{\gamma\delta\lambda}S_{\lambda\beta}$, and $-[\hat{J}_\delta, S_{\gamma\alpha}] = -i\hbar\epsilon_{\delta\gamma\lambda}S_{\lambda\alpha} = +i\hbar\epsilon_{\gamma\delta\lambda}S_{\lambda\alpha}$ were collected under a common $\epsilon_{\gamma\delta\lambda}$. Insert $\hat{J} = \mathbf{S}\hat{J}^{\rm BF}$ in each term and apply (4.6b):

$$\epsilon_{\gamma\delta\lambda}S_{\gamma\alpha}S_{\delta\beta}\,\hat{J}_\lambda = \epsilon_{\alpha\beta\nu}\hat{J}^{\rm BF}_\nu, \quad \epsilon_{\gamma\delta\lambda}S_{\gamma\alpha}S_{\lambda\beta}\,\hat{J}_\delta = \epsilon_{\alpha\nu\beta}\hat{J}^{\rm BF}_\nu, \quad \epsilon_{\gamma\delta\lambda}S_{\delta\beta}S_{\lambda\alpha}\,\hat{J}_\gamma = \epsilon_{\nu\beta\alpha}\hat{J}^{\rm BF}_\nu.$$

Since $\epsilon_{\alpha\beta\nu} + \epsilon_{\alpha\nu\beta} + \epsilon_{\nu\beta\alpha} = \epsilon_{\alpha\beta\nu} - \epsilon_{\alpha\beta\nu} - \epsilon_{\alpha\beta\nu} = -\epsilon_{\alpha\beta\nu}$, the three pieces sum to a single term of *reversed* sign — the extra contribution from differentiating $\mathbf{S}$ outweighs and flips the bare SF algebra:

$$\boxed{[\hat{J}^{\rm BF}_{\alpha}, \hat{J}^{\rm BF}_{\beta}] \;=\; -i\hbar\,\epsilon_{\alpha\beta\gamma}\,\hat{J}^{\rm BF}_{\gamma}.} \tag{4.7}$$

This is the famous **anomalous commutator** for BF angular momentum (Klein, Van Vleck, 1929; reviewed in Bunker–Jensen).

In contrast, BF electronic angular momentum is built intrinsically from BF Cartesian electron operators:

$$[\hat{L}_{{\rm elec},\alpha}, \hat{L}_{{\rm elec},\beta}] \;=\; +i\hbar\,\epsilon_{\alpha\beta\gamma}\,\hat{L}_{{\rm elec},\gamma}, \tag{4.8}$$

the *normal* algebra. Since $\hat{J}^{\rm BF}$ acts only on Euler angles and $\hat{\mathbf{L}}_{\rm elec}$ acts only on BF electron coordinates, the two commute:

$$[\hat{J}^{\rm BF}_{\alpha}, \hat{L}_{{\rm elec},\beta}] \;=\; 0. \tag{4.9}$$

The vibrational $\hat{\pi}_\alpha$, built from Q_b and $\hat{P}_b$, also commutes with both $\hat{J}^{\rm BF}$ and $\hat{\mathbf{L}}_{\rm elec}$. From here on we write $\hat{J}_\alpha \equiv \hat{J}^{\rm BF}_\alpha$ and the anomalous algebra (4.7) is understood.

4.4 Watson pseudopotential

Apply the Podolsky prescription (4.3) to the Watson Hamiltonian (3.22) sector by sector, followed by the rescaling (4.5).

Rotational sector is self-adjoint as written. The kernel $\frac{1}{2}\sum_{\alpha\beta}\hat{A}_\alpha\,\mu_{\alpha\beta}\,\hat{A}_\beta,\ \hat{A}_\alpha \equiv \hat{J}_\alpha - \hat{L}_{\mathrm{elec},\alpha} - \hat{\pi}_\alpha$, contains $\mu_{\alpha\beta}(Q)$, which commutes with $\hat{J}$ (Euler-angle vs. Q sectors) and with $\hat{\mathbf{L}}_{\mathrm{elec}}$ (BF electron coordinates). Since $\mu_{\alpha\beta} = \mu_{\beta\alpha}$,

$$\big(\hat{A}_\alpha\mu_{\alpha\beta}\hat{A}_\beta\big)^\dagger = \hat{A}_\beta\,\mu_{\alpha\beta}\,\hat{A}_\alpha = \hat{A}_\alpha\,\mu_{\beta\alpha}\,\hat{A}_\beta = \hat{A}_\alpha\,\mu_{\alpha\beta}\,\hat{A}_\beta,$$

so the rotational kernel is Hermitian on $\sin\theta\,d\Omega$ and contributes *no* extrapotential; the anomalous algebra (4.7) merely fixes the ordering of the $\hat{A}_\alpha\hat{A}_\beta$ and is absorbed once the symmetric form is adopted.

Vibrational sector generates the pseudopotential. In normal coordinates the vibrational metric is flat ($g^{ab} = \delta_{ab}$) but the volume element carries $\gamma(Q) \equiv \det\mathbf{I}'(Q)$ from (4.4). The Podolsky vibrational operator is therefore

$$\hat{T}_{\mathrm{vib}} = -\frac{\hbar^2}{2}\,\gamma^{-1/2}\sum_b \partial_{Q_b}\big(\gamma^{1/2}\,\partial_{Q_b}\big). \tag{4.10a}$$

The rescaling $\psi_W = \gamma^{1/4}\psi$ of (4.5) sends $\hat{T}_{\mathrm{vib}} \to \gamma^{1/4}\hat{T}_{\mathrm{vib}}\gamma^{-1/4}$. Expanding (the $\frac{1}{4}(\partial_{Q_b}\ln\gamma)\partial_{Q_b}$ cross terms cancel),

$$\begin{aligned}\gamma^{1/4}\hat{T}_{\mathrm{vib}}\gamma^{-1/4} &= -\frac{\hbar^2}{2}\sum_b\Big[\partial_{Q_b}^2 - \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)^2 - \tfrac{1}{4}\partial_{Q_b}^2\ln\gamma\Big] \\ &= \tfrac{1}{2}\sum_b \hat{P}_b^2 + U_{\mathrm{vib}}, \qquad \hat{P}_b = -i\hbar\,\partial_{Q_b},\end{aligned} \tag{4.10b}$$

where $\hat{P}_b$ is now Hermitian on the flat measure $\prod_b dQ_b$ and the extrapotential is the compact "Pauli" form

$$U_{\mathrm{vib}} = \frac{\hbar^2}{8}\sum_b\Big[\partial_{Q_b}^2\ln\gamma + \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)^2\Big] = \frac{\hbar^2}{2}\,\gamma^{-1/4}\Big(\sum_b\partial_{Q_b}^2\Big)\gamma^{1/4}. \tag{4.10c}$$

Collecting the Hermitian rotational kernel, the rescaled vibrational term, and the residue,

$$\begin{aligned}\hat{T}_{\mathrm{rovib}} \ &= \tfrac{1}{2}\sum_{\alpha\beta}\big(\hat{J}_\alpha - \hat{L}_{\mathrm{elec},\alpha} - \hat{\pi}_\alpha\big)\mu_{\alpha\beta}(Q)\big(\hat{J}_\beta - \hat{L}_{\mathrm{elec},\beta} - \hat{\pi}_\beta\big) \\ &\quad + \tfrac{1}{2}\sum_k \hat{P}_b^2 \ + \ \hat{U}(Q).\end{aligned} \tag{4.10}$$

Watson collapse to the trace term. The residue U_{vib} of (4.10c), together with the cross-sector pieces of the Podolsky operator and the reordering of $\hat{\pi}$ against $\mu(Q)$, simplifies dramatically. Because $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^\mathsf{T}$ is *quadratic* in the coordinates through the linear Coriolis vectors $\boldsymbol{\sigma}_b = \sum_a \boldsymbol{\zeta}_{ab}Q_a$ of (3.17a), Jacobi's formula $\partial_{Q_b}\ln\gamma = \sum_{\alpha\beta}\mu_{\alpha\beta}\,\partial_{Q_b}\mathbf{I}'_{\alpha\beta}$ expresses every derivative through the *constant* Coriolis matrices $\boldsymbol{\zeta}_{ab} = \sum_j \mathbf{l}_{ja}\times\mathbf{l}_{jb}$; their contraction sum rules then make all Q-derivative terms cancel against one another, and the total reduces to the celebrated compact **Watson trace** (Watson, 1968; the full reduction is carried out in Appendix B):

$$\boxed{\hat{U}(Q) \;=\; -\frac{\hbar^2}{8}\sum_{\alpha=1}^{3}\mu_{\alpha\alpha}(Q), \qquad \boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})^{-1}.} \tag{4.11}$$

The electrons do not modify $\hat{U}$: $\mu_{\alpha\beta}$ depends on Q only, the BF-electronic sector enters the rovib block solely through $\hat{\mathbf{L}}_{\text{elec}}$ (which commutes with both μ and the Q-measure γ), and the electronic Cartesian Jacobian is flat. The pseudopotential is a c-number on the nuclear configuration space, fixed entirely by the modified inertia $\mathbf{I}'$.

4.4.1 Algebraic cross-check of the coefficient

The $-\hbar^2/8$ is confirmed by the equivalent algebraic (Weyl-ordering) route noted under (4.3): with the Hermitian momenta $\hat{\pi}_A = -i\hbar(\partial_A + \frac{1}{2}\partial_A \ln\sqrt{|g|})$ one has $\hat{T} = \frac{1}{2}\hat{\pi}_A g^{AB}\hat{\pi}_B$, identical to the Laplace–Beltrami operator. Three observations fix the result:

- *Vibrational sector.* With $g^{ab} = \delta_{ab}$ and $\sqrt{|g|} \propto \gamma^{1/2}$, expanding $\frac{1}{2}\sum_b \hat{\pi}^2_{Q_b}$ (the Hermitian momenta of (4.3) specialized to the modes) and rescaling reproduces exactly the residue (4.10c), $U_{\text{vib}} = \frac{\hbar^2}{2}\gamma^{-1/4}\big(\sum_b \partial^2_{Q_b}\big)\gamma^{1/4}$ — the $-\hbar^2/8$ scale enters through the two $\frac{1}{2}\partial_A \ln\sqrt{|g|}$ factors, $\frac{1}{2}\hat{\pi}^2_{Q_b} \supset \frac{\hbar^2}{8}(\partial_{Q_b}\ln\gamma)^2$ etc.
- *Rotational sector.* $\mu_{\alpha\beta}(Q)$ commutes with $\hat{J}$, so the only reordering is among the $\hat{J}$'s; by the anomalous algebra (4.7) the antisymmetric part of $\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta$ vanishes ($\mu_{\alpha\beta} = \mu_{\beta\alpha}$), so this block adds no independent c-number.
- *Reduction.* Carrying the Watson cancellations (Jacobi's formula and the Coriolis sum rules) through both sectors collapses every Q-derivative to the single trace $\hat{U} = -\frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}$, in agreement with (4.11).

The pseudopotential is a c-number depending only on Q (a function on the nuclear configuration space).

4.4.2 Operator ordering: what is and isn't ambiguous

- $\mu_{\alpha\beta}(Q)$ commutes with $\hat{J}$ (different coordinate sectors: Euler angles vs. Q), with $\hat{\mathbf{L}}_{\text{elec}}$ (BF electron coords), but not with $\hat{P}_b$ (since $\partial\mu_{\alpha\beta}/\partial Q_b \neq 0$). Hence $\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta = \mu_{\alpha\beta}\hat{J}_\alpha\hat{J}_\beta$.
- Within the rotational block, the symmetric product $\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta$ in (4.10) is unambiguous because μ is symmetric in (α, β), and the Watson pseudopotential (4.11) accounts for the residual non-commutativity of the $\hat{J}$'s themselves.
- In normal coordinates the vibrational term is the bare $\frac{1}{2}\sum_b \hat{P}_b^2$; the Q-dependence of the volume element has been moved into $\hat{U}(Q)$ by the rescaling (4.5), which is precisely what Podolsky produces.

4.5 Electronic kinetic operator

In BF Cartesian electron coordinates, $\hat{\bar{\mathbf{p}}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ is Hermitian on $L^2(\mathbb{R}^{3N_{\rm elec}}, d^{3N_{\rm elec}}\bar{\mathbf{r}})$. The electronic + mass-polarization kinetic operator is therefore the straightforward quantization of (3.21d):

$$\hat{T}_e \;=\; \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^2}{2m_e} \;+\; \frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2. \tag{4.12}$$

Expanding the second piece:

$$\frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2 \;=\; \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^2}{2M_N} \;+\; \frac{1}{M_N}\sum_{i<l} \hat{\bar{\mathbf{p}}}_i\cdot\hat{\bar{\mathbf{p}}}_l. \tag{4.13}$$

The diagonal $1/(2M_N)$ piece renormalizes the effective electron mass through $1/m_e + 1/M_N = M_{\rm tot}/(m_e M_N) = 1/\mu_e$. The off-diagonal two-electron term is the mass-polarization *sensu stricto*, dynamically coupling electrons through nuclear recoil — a purely classical effect (Section 2.4) inherited by the quantum theory.

4.6 Complete quantum molecular Hamiltonian

Combining (4.10), (4.11), (4.12), and the rotationally invariant Coulomb potentials from (3.8):

$$\boxed{\begin{aligned}\hat{H}_{\rm int} \;&= \tfrac{1}{2}\sum_{\alpha\beta}\big(\hat{J}_\alpha - \hat{L}_{{\rm elec},\alpha} - \hat{\pi}_\alpha\big)\mu_{\alpha\beta}(Q)\big(\hat{J}_\beta - \hat{L}_{{\rm elec},\beta} - \hat{\pi}_\beta\big)\\ &\quad + \tfrac{1}{2}\sum_k \hat{P}_b^2\\ &\quad + \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^2}{2m_e} \;+\; \frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2\\ &\quad - \frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}(Q)\\ &\quad + V_{NN}(Q) \;+\; V_{Ne}(Q,\{\bar{\mathbf{r}}_i\}) \;+\; V_{ee}(\{\bar{\mathbf{r}}_i\}),\end{aligned}} \tag{4.14}$$

acting on $L^2(\mathcal{C},\, d\mu_W)$ with $d\mu_W = \sin\theta\, d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3\bar{\mathbf{r}}_i$. All operators are Hermitian on this Hilbert space.

This is the **exact non-relativistic molecular Hamiltonian** after separation of the total COM motion, in BF + Eckart frame, with electrons referred to the nuclear COM. The only approximations made along the way were: (i) non-relativistic dynamics; (ii) Eckart's choice of the BF orientation (a convention, not an approximation); (iii) parametrization of the nuclear shape by $3N_{\rm nucl} - 6$ internal coordinates Q_b (a parametrization, also exact).

4.7 Born–Huang expansion

4.7.1 Clamped-nuclei electronic Hamiltonian

At each nuclear configuration Q, define the **BF clamped-nuclei electronic Hamiltonian** as the sum of all Q-parametric electron-only pieces of (4.14):

$$\hat{H}_{\rm el}(Q) \;\equiv\; \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^2}{2m_e} \;+\; \frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2 \;+\; V_{Ne}(Q,\{\bar{\mathbf{r}}_i\}) \;+\; V_{ee}(\{\bar{\mathbf{r}}_i\}). \tag{4.15}$$

Remark. The mass-polarization piece $(\sum_i \hat{\bar{\mathbf{p}}}_i)^2/(2M_N)$ is included here as a small correction with $1/M_N$. Some treatments ("clamped-nuclei BO") drop it; we keep it explicit because it is parametric in Q only through the absent dependence (i.e., M_N is just a number), which means it acts trivially on the parametric family of electronic Hilbert spaces.

Solve the parametric eigenvalue problem:

$$\hat{H}_{\rm el}(Q)\,|\phi_n(Q)\rangle \;=\; E_n^{\rm el}(Q)\,|\phi_n(Q)\rangle, \tag{4.16}$$

in the electronic Hilbert space $\mathcal{H}_e = L^2(\mathbb{R}^{3N_{\rm elec}}, d^{3N_{\rm elec}}\bar{\mathbf{r}})$. The eigenstates $\{|\phi_n(Q)\rangle\}$ depend *parametrically* on Q — Q is treated as a fixed classical parameter, not a quantum degree of freedom — and form a complete orthonormal basis of $\mathcal{H}_e$ at each Q:

$$\sum_n |\phi_n(Q)\rangle\langle\phi_n(Q)| \;=\; \hat{\mathbb{1}}_e, \qquad \langle\phi_m(Q)|\phi_n(Q)\rangle_e = \delta_{mn}. \tag{4.17}$$

4.7.2 The exact Born–Huang expansion

The *exact* molecular wavefunction $|\Psi\rangle \in L^2(\mathcal{C}, d\mu_W)$ admits a resolution against the parametric electronic basis. Insert the resolution of identity (4.17) at each Q:

$$|\Psi\rangle \;=\; \sum_n |\chi_n\rangle \otimes |\phi_n(Q)\rangle, \qquad \chi_n(\Omega,Q) \;\equiv\; \langle\phi_n(Q)|\Psi\rangle_e, \tag{4.18}$$

where $\Omega = (\phi,\theta,\chi)$, the electronic inner product is taken at fixed Q, and χ_n are the **nuclear coefficient functions**. The expansion (4.18) is **exact**, not an ansatz: it follows from the completeness (4.17) at each Q, with the only subtlety being that the basis $|\phi_n(Q)\rangle$ varies parametrically. There is *no* simple tensor factorization $\mathcal{H}_{\rm molecule} = \mathcal{H}_N \otimes \mathcal{H}_e$ because of this parametric dependence; the Born–Huang expansion is the rigorous replacement.

4.7.3 Coupled-channel equations

Substitute (4.18) into $\hat{H}_{\rm int}|\Psi\rangle = E|\Psi\rangle$, project from the left by $\langle\phi_m(Q)|$, and use (4.16):

$$\big[\hat{T}_N + E_n^{\rm el}(Q) + V_{NN}(Q) + \hat{U}(Q)\big]\chi_n(\Omega,Q) \;+\; \sum_m \hat{\Lambda}_{nm}\,\chi_m(\Omega,Q) \;=\; E\,\chi_n(\Omega,Q), \tag{4.19}$$

where $\hat{T}_N$ denotes the nuclear (rotational + vibrational) kinetic operator from the first two lines of (4.14). The **non-adiabatic coupling operators** $\hat{\Lambda}_{nm}$ arise from the action of $\hat{T}_N$ (and the rot–electronic couplings through $\hat{\mathbf{L}}_{\rm elec}$) on the Q-parametric dependence of

$|\phi_m(Q)\rangle$:

$$\hat{\Lambda}_{nm} \;\sim\; -\frac{\hbar^2}{1}\,\langle\phi_n|\nabla_Q\phi_m\rangle_e\cdot\nabla_Q \;-\; \frac{\hbar^2}{2}\,\langle\phi_n|\nabla_Q^2\phi_m\rangle_e \;+\; \dots, \tag{4.20}$$

with analogous terms from the rotation–electron Coriolis $\hat{J}_\alpha\hat{L}_{\mathrm{elec},\alpha}$ kernel.

4.7.4 Born–Oppenheimer approximation as truncation

The **Born–Oppenheimer (BO) approximation** is the diagonal truncation of (4.19): set $\hat{\Lambda}_{nm} \to 0$ for $m \neq n$ and (optionally) also $\hat{\Lambda}_{nn} \to 0$. The resulting effective nuclear problem is

$$\big[\hat{T}_N + E_n^{\mathrm{el}}(Q) + V_{NN}(Q) + \hat{U}(Q)\big]\chi_n^{\mathrm{BO}}(\Omega, Q) \;=\; E^{\mathrm{BO}}\,\chi_n^{\mathrm{BO}}(\Omega, Q), \tag{4.21}$$

where the **Born–Oppenheimer potential energy surface** for electronic state n is

$$V_n^{\mathrm{BO}}(Q) \;\equiv\; E_n^{\mathrm{el}}(Q) + V_{NN}(Q) + \hat{U}(Q). \tag{4.22}$$

The exactness of (4.18) and the well-definedness of (4.19) are mathematical facts. BO is the physical assumption that off-diagonal $\hat{\Lambda}_{nm}$ are small — justified when electronic gaps are large compared to nuclear kinetic energies, breaking down at conical intersections and avoided crossings, where the parametric variation of $|\phi_n(Q)\rangle$ becomes singular.

4.7.5 Summary

1. (4.14) is the exact non-relativistic quantum molecular Hamiltonian in BF + Eckart + nuclear-COM coordinates.

2. The Born–Huang expansion (4.18) is exact, following from completeness of the parametric electronic basis (4.17).

3. The infinite coupled system (4.19) is equivalent to the Schrödinger equation $\hat{H}_{\mathrm{int}}|\Psi\rangle = E|\Psi\rangle$.

4. The Born–Oppenheimer truncation reduces it to (4.21), giving the familiar single-PES picture (4.22).

Appendix

Appendix A: Mass-Polarization Term in Classical Molecular Hamiltonian

We start from the laboratory-frame kinetic energy

$$T = \frac{1}{2}\sum_j M_j \dot{\mathbf{R}}_j^2 + \frac{1}{2}\sum_i m_e \dot{\mathbf{r}}_i^2, \tag{1}$$

where M_j is the mass of the j-th nucleus, $\mathbf{R}_j$ is its Cartesian coordinate, m_e is the electron mass, and $\mathbf{r}_i$ is the coordinate of the i-th electron.

4.7.6 Nuclear Center-of-Mass Coordinate

Define the nuclear total mass

$$M_N = \sum_j M_j. \tag{2}$$

The nuclear center-of-mass coordinate is

$$\mathbf{R}_{\mathrm{cm}} = \mathbf{R}_N^{\mathrm{CoM}} = \frac{1}{M_N}\sum_j M_j \mathbf{R}_j. \tag{3}$$

We introduce coordinates shifted by the nuclear center of mass:

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\mathrm{cm}}, \tag{4}$$

$$\mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\mathrm{cm}}. \tag{5}$$

Therefore,

$$\mathbf{R}_j = \mathbf{R}_{\mathrm{cm}} + \mathbf{R}'_j, \qquad \mathbf{r}_i = \mathbf{R}_{\mathrm{cm}} + \mathbf{r}'_i. \tag{6}$$

The shifted nuclear coordinates satisfy the constraint

$$\sum_j M_j \mathbf{R}'_j = 0, \tag{7}$$

and therefore

$$\sum_j M_j \dot{\mathbf{R}}'_j = 0. \tag{8}$$

This is important: the $\mathbf{R}'_j$ are not all independent.

4.7.7 Kinetic Energy in Velocity Form

Substitute

$$\dot{\mathbf{R}}_j = \dot{\mathbf{R}}_{\mathrm{cm}} + \dot{\mathbf{R}}'_j, \qquad \dot{\mathbf{r}}_i = \dot{\mathbf{R}}_{\mathrm{cm}} + \dot{\mathbf{r}}'_i \tag{9}$$

into the kinetic energy. For the nuclear part,

$$T_N = \frac{1}{2}\sum_j M_j \left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{R}}'_j\right)^2 \tag{10}$$

$$= \frac{1}{2}M_N\dot{\mathbf{R}}^2_{\text{cm}} + \dot{\mathbf{R}}_{\text{cm}} \cdot \sum_j M_j\dot{\mathbf{R}}'_j + \frac{1}{2}\sum_j M_j\dot{\mathbf{R}}'^2_j. \tag{11}$$

Using

$$\sum_j M_j\dot{\mathbf{R}}'_j = 0, \tag{12}$$

we get

$$T_N = \frac{1}{2}M_N\dot{\mathbf{R}}^2_{\text{cm}} + \frac{1}{2}\sum_j M_j\dot{\mathbf{R}}'^2_j. \tag{13}$$

For the electronic part,

$$T_e = \frac{1}{2}\sum_i m_e \left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}'_i\right)^2 \tag{14}$$

$$= \frac{1}{2}N_{\text{elec}}m_e\dot{\mathbf{R}}^2_{\text{cm}} + m_e\dot{\mathbf{R}}_{\text{cm}} \cdot \sum_i \dot{\mathbf{r}}'_i + \frac{1}{2}\sum_i m_e\dot{\mathbf{r}}'^2_i, \tag{15}$$

where N_{elec} is the number of electrons. Hence the total kinetic energy is

$$T = \frac{1}{2}\left(M_N + N_{\text{elec}}m_e\right)\dot{\mathbf{R}}^2_{\text{cm}} + m_e\dot{\mathbf{R}}_{\text{cm}} \cdot \sum_i \dot{\mathbf{r}}'_i + \frac{1}{2}\sum_j M_j\dot{\mathbf{R}}'^2_j + \frac{1}{2}\sum_i m_e\dot{\mathbf{r}}'^2_i. \tag{16}$$

4.7.8 Canonical Momenta

The momentum conjugate to the nuclear center-of-mass coordinate $\mathbf{R}_{\text{cm}}$ is

$$\mathbf{P}_{\text{cm}} = \frac{\partial\mathcal{L}}{\partial\dot{\mathbf{R}}_{\text{cm}}} \tag{17}$$

$$= \left(M_N + N_{\text{elec}}m_e\right)\dot{\mathbf{R}}_{\text{cm}} + m_e\sum_i \dot{\mathbf{r}}'_i. \tag{18}$$

Since

$$\mathbf{p}'_i = \frac{\partial\mathcal{L}}{\partial\dot{\mathbf{r}}'_i} = m_e\left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}'_i\right), \tag{19}$$

we can write

$$\sum_i \mathbf{p}'_i = N_{\text{elec}}m_e\dot{\mathbf{R}}_{\text{cm}} + m_e\sum_i \dot{\mathbf{r}}'_i. \tag{20}$$

Therefore,

$$\boxed{\mathbf{P}_{\text{cm}} = M_N\dot{\mathbf{R}}_{\text{cm}} + \sum_i \mathbf{p}'_i} \tag{21}$$

or equivalently,

$$\boxed{M_N\dot{\mathbf{R}}_{\text{cm}} = \mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i} \tag{22}$$

and hence

$$\boxed{\dot{\mathbf{R}}_{\rm cm} = \frac{1}{M_N}\left(\mathbf{P}_{\rm cm} - \sum_i \mathbf{p}'_i\right).} \tag{23}$$

The electron momentum conjugate to $\mathbf{r}'_i$ is

$$\boxed{\mathbf{p}'_i = m_e\left(\dot{\mathbf{R}}_{\rm cm} + \dot{\mathbf{r}}'_i\right).} \tag{24}$$

Thus $\mathbf{p}'_i$ is the lab-frame electron momentum, not simply $m_e\dot{\mathbf{r}}'_i$. Equivalently,

$$m_e\dot{\mathbf{r}}'_i = \mathbf{p}'_i - m_e\dot{\mathbf{R}}_{\rm cm}. \tag{25}$$

The nuclear shifted momenta are

$$\boxed{\mathbf{P}'_j = M_j\dot{\mathbf{R}}'_j.} \tag{26}$$

Because the shifted nuclear coordinates satisfy

$$\sum_j M_j\mathbf{R}'_j = 0, \tag{27}$$

their conjugate momenta satisfy

$$\boxed{\sum_j \mathbf{P}'_j = 0.} \tag{28}$$

In terms of the laboratory-frame nuclear momenta

$$\mathbf{P}_j = M_j\dot{\mathbf{R}}_j, \tag{29}$$

one has

$$\boxed{\mathbf{P}'_j = \mathbf{P}_j - \frac{M_j}{M_N}\mathbf{P}_N,} \tag{30}$$

where

$$\mathbf{P}_N = \sum_j \mathbf{P}_j = M_N\dot{\mathbf{R}}_{\rm cm}. \tag{31}$$

Using

$$\mathbf{P}_N = \mathbf{P}_{\rm cm} - \sum_i \mathbf{p}'_i, \tag{32}$$

we may also write

$$\boxed{\mathbf{P}'_j = \mathbf{P}_j - \frac{M_j}{M_N}\left(\mathbf{P}_{\rm cm} - \sum_i \mathbf{p}'_i\right).} \tag{33}$$

4.7.9 Kinetic Energy in Momentum Form

The original kinetic energy can be written as

$$T = T_{\rm nuc,int} + T_{\rm nuc,CoM} + T_e. \tag{34}$$

The nuclear internal kinetic energy is

$$T_{\text{nuc,int}} = \sum_j \frac{\mathbf{P}'^2_j}{2M_j}. \tag{35}$$

The electron kinetic energy is

$$T_e = \sum_i \frac{\mathbf{p}'^2_i}{2m_e}. \tag{36}$$

The nuclear center-of-mass kinetic energy is

$$T_{\text{nuc,CoM}} = \frac{1}{2} M_N \dot{\mathbf{R}}^2_{\text{cm}}. \tag{37}$$

Using

$$M_N \dot{\mathbf{R}}_{\text{cm}} = \mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i, \tag{38}$$

we get

$$T_{\text{nuc,CoM}} = \frac{1}{2M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i \right)^2. \tag{39}$$

Therefore the kinetic energy in terms of the new momenta is

$$\boxed{T = \sum_j \frac{\mathbf{P}'^2_j}{2M_j} + \sum_i \frac{\mathbf{p}'^2_i}{2m_e} + \frac{1}{2M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i \right)^2.} \tag{40}$$

The accompanying constraints are

$$\boxed{\sum_j M_j \mathbf{R}'_j = 0, \qquad \sum_j \mathbf{P}'_j = 0.} \tag{41}$$

4.7.10 Expanded Form

Expanding the center-of-mass recoil term gives

$$T = \sum_j \frac{\mathbf{P}'^2_j}{2M_j} + \sum_i \frac{\mathbf{p}'^2_i}{2m_e} + \frac{\mathbf{P}^2_{\text{cm}}}{2M_N} - \frac{\mathbf{P}_{\text{cm}}}{M_N} \cdot \sum_i \mathbf{p}'_i + \frac{1}{2M_N} \left(\sum_i \mathbf{p}'_i \right)^2. \tag{42}$$

Thus,

$$\boxed{T = \sum_j \frac{\mathbf{P}'^2_j}{2M_j} + \sum_i \frac{\mathbf{p}'^2_i}{2m_e} + \frac{\mathbf{P}^2_{\text{cm}}}{2M_N} - \frac{\mathbf{P}_{\text{cm}}}{M_N} \cdot \sum_i \mathbf{p}'_i + \frac{1}{2M_N} \left(\sum_i \mathbf{p}'_i \right)^2.} \tag{43}$$

4.7.11 Hamiltonian

If the potential energy is translationally invariant, then

$$V = V\left(\{\mathbf{R}'_j\}, \{\mathbf{r}'_i\}\right). \tag{44}$$

The Hamiltonian is therefore

$$\boxed{H = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N}\left(\mathbf{P}_{\rm cm} - \sum_i \mathbf{p}_i'\right)^2 + V\left(\{\mathbf{R}_j'\}, \{\mathbf{r}_i'\}\right).} \tag{45}$$

4.7.12 Molecular Rest Frame

If the whole molecule is taken to be in its rest frame, then

$$\mathbf{P}_{\rm cm} = 0. \tag{46}$$

The kinetic energy becomes

$$\boxed{T_{\rm rest} = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N}\left(\sum_i \mathbf{p}_i'\right)^2.} \tag{47}$$

The last term,

$$\frac{1}{2M_N}\left(\sum_i \mathbf{p}_i'\right)^2, \tag{48}$$

is the nuclear center-of-mass recoil term, also called the mass-polarization term.

4.7.13 Important Interpretation

The coordinate transformation used here shifts both nuclei and electrons by the nuclear center of mass, not by the total molecular center of mass. Therefore,

$$\mathbf{p}_i' = m_e\left(\dot{\mathbf{R}}_{\rm cm} + \dot{\mathbf{r}}_i'\right) \tag{49}$$

is the canonical momentum conjugate to $\mathbf{r}_i'$. It is equal to the electron lab-frame mechanical momentum. However,

$$m_e\dot{\mathbf{r}}_i' \tag{50}$$

is only the electron momentum relative to the moving nuclear center of mass. It is not the canonical momentum conjugate to $\mathbf{r}_i'$ unless the nuclear center-of-mass motion has been fixed or eliminated. The final kinetic energy in nuclear-CoM coordinates is therefore

$$\boxed{T = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N}\left(\mathbf{P}_{\rm cm} - \sum_i \mathbf{p}_i'\right)^2,} \tag{51}$$

with

$$\boxed{\sum_j M_j\mathbf{R}_j' = 0, \qquad \sum_j \mathbf{P}_j' = 0.} \tag{52}$$

Appendix B: Full Derivation of the Watson Pseudopotential

Here we supply the reduction quoted between (4.10c) and (4.11): how curvilinear quantization of the Watson Hamiltonian (3.22), after the rescaling (4.5), produces the compact pseudopotential $\hat{U} = -\frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}$.

B.1 The general curvilinear extrapotential

Let q^A be arbitrary coordinates with metric g_{AB}, determinant $g = \det g_{AB}$, and classical kinetic energy $2T = g^{AB}p_Ap_B$. The unique kinetic operator Hermitian on $L^2(\sqrt{g}\,d^nq)$ is the Laplace–Beltrami operator

$$\hat{T}_{\rm LB} = -\frac{\hbar^2}{2}\,g^{-1/2}\,\partial_A\big(g^{1/2}g^{AB}\partial_B\big). \tag{B.1}$$

To pass to a flat measure $\prod_A dq^A$ we rescale $\psi_W = g^{1/4}\psi$, i.e. $\hat{T}_W = g^{1/4}\hat{T}_{\rm LB}g^{-1/4}$. With $\rho \equiv g^{1/2}$ and $f_B \equiv \partial_B\psi_W - \frac{1}{2}(\partial_B\ln\rho)\psi_W$,

$$\hat{T}_W\psi_W = -\frac{\hbar^2}{2}\Big[\tfrac{1}{2}(\partial_A\ln\rho)\,g^{AB}f_B + (\partial_Ag^{AB})f_B + g^{AB}\partial_Af_B\Big]. \tag{B.2}$$

The terms linear in $\partial\psi_W$ are $\frac{1}{2}g^{AB}(\partial_A\ln\rho)\partial_B\psi_W$ (first bracket) and $-\frac{1}{2}g^{AB}(\partial_B\ln\rho)\partial_A\psi_W$ (third bracket); by symmetry of g^{AB} they cancel, while the second-derivative part re-assembles into the flat symmetric operator $\frac{1}{2}\hat{p}_Ag^{AB}\hat{p}_B$ ($\hat{p}_A = -i\hbar\partial_A$). The surviving ψ_W-proportional pieces define the *extrapotential*; using $\ln\rho = \frac{1}{2}\ln g$,

$$\boxed{\hat{T}_W = \tfrac{1}{2}\,\hat{p}_A\,g^{AB}\,\hat{p}_B + U, \qquad U = \frac{\hbar^2}{8}\Big[\tfrac{1}{4}\,g^{AB}\,\partial_A\ln g\,\partial_B\ln g + \partial_A\big(g^{AB}\partial_B\ln g\big)\Big].} \tag{B.3}$$

Equation (B.3) is exact and coordinate-general; the pseudopotential is the entire c-number U, and the prefactor $\frac{\hbar^2}{8}$ traces to the two half-powers $g^{\pm1/4}$ of the rescaling.

B.2 The molecular metric

For (3.22) the coordinates are $q^A = (\theta, \phi, \chi, Q_b)$. By (4.4), $\sqrt{|g|} = \sin\theta\,(\det\mathbf{I}')^{1/2}$, so

$$\ln g = 2\ln\sin\theta + \ln\gamma, \qquad \gamma \equiv \det\mathbf{I}'(Q) = 1/\det\boldsymbol{\mu}, \tag{B.4}$$

and all Q-dependence of the measure sits in γ. Reading the contravariant metric off (3.22) in the body-frame momentum basis $(\hat{J}_\alpha, \hat{P}_b)$ [with $\hat{\pi}_\alpha = \sum_b\sigma_{b\alpha}\hat{P}_b$, $\sigma_{b\alpha} = (\boldsymbol{\sigma}_b)_\alpha$],

$$\mathsf{g}^{-1} = \begin{pmatrix}\boldsymbol{\mu} & -\boldsymbol{\mu}\boldsymbol{\sigma}^{\mathsf{T}}\\ -\boldsymbol{\sigma}\boldsymbol{\mu} & \mathbb{1}+\boldsymbol{\sigma}\boldsymbol{\mu}\boldsymbol{\sigma}^{\mathsf{T}}\end{pmatrix}, \qquad \det\mathsf{g}^{-1} = \det\boldsymbol{\mu} = 1/\gamma, \tag{B.5}$$

the determinant following from the Schur reduction $\det(\mathbb{1}+\boldsymbol{\sigma}\boldsymbol{\mu}\boldsymbol{\sigma}^{\mathsf{T}} - \boldsymbol{\sigma}\boldsymbol{\mu}\,\boldsymbol{\mu}^{-1}\boldsymbol{\mu}\boldsymbol{\sigma}^{\mathsf{T}}) = 1$, consistent with (B.4).

B.3 Reduction to the trace

In the rovib convention the body-frame angular momenta $\hat{J}_\alpha$ already carry $\sin\theta$ (they are Hermitian on $\sin\theta\,d\Omega$, with the anomalous algebra (4.7)), so only the residual weight $\gamma^{1/2}(Q)$ is removed by the rescaling (4.5). Three groups of terms arise.

(i) Rotational. In $\frac{1}{2}\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta$ the factor $\mu_{\alpha\beta}(Q)$ commutes with $\hat{J}$ and is symmetric; writing $\frac{1}{2}\mu_{\alpha\beta}\hat{J}_\alpha\hat{J}_\beta = \frac{1}{4}\mu_{\alpha\beta}\{\hat{J}_\alpha, \hat{J}_\beta\} + \frac{1}{4}\mu_{\alpha\beta}[\hat{J}_\alpha, \hat{J}_\beta]$ and using the anomalous $[\hat{J}_\alpha, \hat{J}_\beta] =$

$-i\hbar\epsilon_{\alpha\beta\gamma}\hat{J}_\gamma$ with $\mu_{\alpha\beta}\epsilon_{\alpha\beta\gamma} = 0$, the commutator drops: the asymmetric-top operator is Hermitian as written and yields *no* c-number.

(ii) Vibrational. The diagonal $\hat{P}_b^2$ part, with weight $\gamma^{1/2}$, gives $\frac{1}{2}\sum_b \hat{P}_b^2$ plus the Pauli residue [exactly (4.10c)],

$$U_{\text{vib}} = \frac{\hbar^2}{2}\,\gamma^{-1/4}\Big(\sum_b \partial_b^2\Big)\gamma^{1/4} = \frac{\hbar^2}{8}\sum_b\Big[\partial_b^2\ln\gamma + \tfrac{1}{4}(\partial_b\ln\gamma)^2\Big]. \tag{B.6}$$

(iii) Coriolis. The off-diagonal $-\hat{J}_\alpha(\boldsymbol{\mu}\boldsymbol{\sigma}^{\mathsf{T}})_{\alpha b}\hat{P}_b$ block, together with the γ-dressing of $\hat{P}_b$ and the commutators $[\hat{P}_b, \mu_{\alpha\beta}]$ and $[\hat{P}_b, \sigma_{a\alpha}]$, supplies the remaining Q-derivative c-numbers.

The collapse rests on two exact identities. Jacobi's formula for $\gamma = \det\mathbf{I}'$,

$$\partial_b\ln\gamma = \operatorname{tr}\!\big(\boldsymbol{\mu}\,\partial_b\mathbf{I}'\big), \qquad \partial_b\boldsymbol{\mu} = -\,\boldsymbol{\mu}\,(\partial_b\mathbf{I}')\,\boldsymbol{\mu}, \tag{B.7}$$

rewrites every $\partial_b\ln\gamma$ and $\partial_b\boldsymbol{\mu}$ appearing in (B.6) and (iii) through $\partial_b\mathbf{I}'$; and because $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}$ with $\boldsymbol{\sigma}_b = \sum_a \boldsymbol{\zeta}_{ab}Q_a$ of (3.17a) *linear* in Q, the derivatives $\partial_b\mathbf{I}'$ are built from the *constant* Coriolis matrices $\boldsymbol{\zeta}_{ab}$, whose contractions satisfy the orthogonality sum rules of the Eckart mode vectors ($\sum_j \mathbf{l}_{ja}\cdot\mathbf{l}_{jb} = \delta_{ab}$; the $\boldsymbol{\zeta}^{(\alpha)}$ are the antisymmetric $SO(3)$ generators on the mode index). Substituting (B.7) into groups (ii) and (iii), the gradient terms $\propto \partial_b\operatorname{tr}\boldsymbol{\mu}$ produced by the vibrational residue are cancelled identically by those of the Coriolis block, and the surviving Q-local remainder is the single trace

$$\boxed{\hat{U} \;=\; U_{\text{rot}} + U_{\text{vib}} + U_{\text{Cor}} \;=\; -\frac{\hbar^2}{8}\sum_{\alpha=1}^{3}\mu_{\alpha\alpha}(Q)} \tag{B.8}$$

of (4.11). This reproduces Watson's reduction [J. K. G. Watson, *Mol. Phys.* **15**, 479 (1968)]. The cancellation is special to rectilinear normal coordinates: for general curvilinear Q_b the residue (B.6) does *not* collapse to the trace, leaving the additional vibrational extrapotential terms anticipated in Section 4.1. The electronic sector contributes nothing to $\hat{U}$, since $\boldsymbol{\mu}$, γ, and $\boldsymbol{\sigma}$ depend on Q alone.

References